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OVERCURRENT PROTECTION FOR A LOAD CONTROL DEVICE

Abstract

A load control device may include a closed-loop gate drive circuit coupled to a semiconductor switch and configured to render the semiconductor switch conductive or non-conductive at a control time during half-cycles of an AC mains line voltage. The gate drive circuit may limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit. The load control device may include a control circuit that is configured to set the maximum current limit to a second magnitude for a period of time following a control time of the semiconductor switch during a conduction period of a half-cycle of the AC mains line voltage, and set the maximum current limit to a first magnitude after the expiration of the period of time during the conduction period of the half-cycle of the AC mains line voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority from Provisional U.S. Patent Application No. 63/560,212, filed Mar. 1, 2024, the entire disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] Prior art load control devices, such as dimmer switches, may be coupled in series electrical connection between an alternating-current (AC) power source and a lighting load for controlling the amount of power delivered from the AC power source to the lighting load. A standard dimmer switch may typically comprise a bidirectional semiconductor switch, e.g., a thyristor (e.g., such as a triac) or two field-effect transistors (FETs) in anti-series connection. The bidirectional semiconductor switch may be coupled in series between the AC power source and the load. The bidirectional semiconductor switch may be controlled to be conductive and non-conductive for portions of a half-cycle of the AC power source to thus control the amount of power delivered to the electrical load.

[0003] Generally, dimmer switches may use either a forward phase-control dimming technique or a reverse phase-control dimming technique in order to control when the bidirectional semiconductor switch is rendered conductive and non-conductive to thus control the power delivered to the load. The dimmer switch may comprise a toggle actuator for turning the lighting load on and off and an intensity adjustment actuator for adjusting the intensity level of the lighting load. Examples of prior art dimmer switches are described in greater detail is commonly-assigned U.S. Pat. No. 5,248,919, issued Sep. 29, 1993, entitled LIGHTING CONTROL DEVICE; and U.S. Pat. No. 6,969,959, issued Nov. 29, 2005, entitled ELECTRONIC CONTROL SYSTEMS AND METHODS; the entire disclosures of which are incorporated by reference herein.

SUMMARY

[0004] As described herein, a load control device may be configured to control power delivered from an alternating-current (AC) power source to an electrical load. The load control device may include a controllably conductive device adapted to be coupled in series between the AC power source and the electrical load. The AC power source may be configured to generate an AC mains line voltage. The controllably conductive device may include a semiconductor switch configured to conduct a load current through the electrical load. The load control device may include a closed-loop gate drive circuit coupled to the semiconductor switch and configured to render the semiconductor switch conductive or non-conductive at a control time during half-cycles of the AC mains line voltage. The closed-loop gate drive circuit may be configured to limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit. The load control device may include a control circuit that is configured to generate a drive signal to adjust the control time of the semiconductor switch during each half-cycle of the AC mains line voltage to control the amount of power delivered to the electrical load. The control circuit may be configured to control the drive signal to render the semiconductor switch conductive at a control time during a half-cycle of the AC mains line voltage to maintain the semiconductor switch conductive for a conduction period during the half-cycle of the AC mains line voltage. The control circuit may set the maximum current limit of the closed-loop gate drive circuit to a first limit for a period of time following the control time during the conduction period of the half-cycle of the AC mains line voltage. The control circuit may set the maximum current limit of the closed-loop gate drive circuit to a second limit after the expiration of the period of time during the conduction period of the half-cycle of the AC mains line voltage, wherein the second limit is less than the first limit.

[0005] The control circuit may be configured to set the maximum current limit to the first limit for a pulse period that starts at the control time during the half-cycle of the AC mains line voltage, and set the maximum current limit to the second limit at the end of the pulse period and during a remainer of the conduction period of the half-cycle of the AC mains line voltage. The pulse period may be approximately 500 microseconds. The control circuit may be configured to set the maximum current limit to the first limit for the pulse period to allow for a capacitance of a capacitive load to charge during the half-cycle of the AC mains line voltage.
[0006] The control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to the first limit for the pulse period following the control time of the semiconductor switch during an amount of time (e.g., an initial plurality of half-cycles of the AC mains line voltage) after turning on the electrical load, and after the initial plurality of half-cycles of the AC mains line voltage, set the maximum current limit of the closed-loop gate drive circuit to the second limit at the control time during subsequent half-cycles of the AC mains line voltage after the initial plurality of half-cycles and while controlling power to the electrical load. The control circuit may be configured to shorten a duration of the pulse period over the initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load.
[0007] The control circuit is configured to set a length of the pulse period to a second time period for a number of half-cycles after turning on the electrical load, and set the length of the pulse period to a first time period after the number of half-cycles and while controlling power to the electrical load. The first time period may be approximately 500 microseconds, the second time period may be approximately 2,000 microseconds, and/or the number of half-cycles may be approximately 120 half-cycles.
[0008] The control circuit may be configured to adjust the maximum current limit across a range of values from the first limit to the second limit over a plurality of half-cycles of the AC mains line voltage. The control circuit may be configured to linearly reduce the maximum current limit across the range of values from the first limit to the second limit over the plurality of half-cycles of the AC mains line voltage.
[0009] When operating in a forward phase-control technique, the control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to the first limit for the period of time following the control time of the semiconductor switch during a conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load. The control circuit may set the maximum current limit of the closed-loop gate drive circuit to the second limit after the expiration of the period of time during the conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load. For example, when operating in a reverse phase-control technique, the control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to a third limit during the entire conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load. The third limit may be equal to the first limit.
[0010] The closed-loop gate drive circuit may be configured to generate a target signal in response to the control circuit, receive a feedback signal indicative of a magnitude of the load current conducted through the semiconductor switch, and generate a gate control signal in response to the target signal and the feedback signal. The semiconductor switch of the controllably conductive device may be configured to be rendered conductive and non-conductive in response to the gate control signal.
[0011] When using a forward phase-control technique, the control circuit may be configured to control the drive signal to render the semiconductor switch non-conductive at a beginning of each half-cycle of the AC mains line voltage, and control the drive signal to render the semiconductor switch conductive at the control time during the each half-cycle of the AC mains line voltage and throughout a remainder of each half-cycle of the AC mains line voltage while controlling power to the electrical load. For example, when using a reverse phase-control technique, the control circuit

may be configured to control the drive signal to render the semiconductor switches conductive at the beginning of each half-cycle of the AC mains line voltage, and control the drive signal to render the semiconductor switch non-conductive at the control time during each half-cycle of the AC mains line voltage.

[0012] The control circuit may be configured to control the semiconductor switch to be non-conductive for a non-conduction period and conductive for a conduction period during one or more half-cycles of the AC mains line voltage to control the amount of power delivered to the electrical load.

[0013] Some examples may include a load control device that is configured to control power delivered from an alternating-current (AC) power source to an electrical load, where the load control device includes a controllably conductive device adapted to be coupled in series between the AC power source and the electrical load, where the AC power source is configured to generate an AC mains line voltage, and where the controllably conductive device comprises a semiconductor switch configured to conduct a load current through the electrical load. The load control device may include a closed-loop gate drive circuit coupled to the semiconductor switch and that is configured to render the semiconductor switch conductive or non-conductive at a control time during half-cycles of the AC mains line voltage. The closed-loop gate drive circuit may be configured to limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit, where the maximum current limit may be adjustable between a first limit and a second limit. The load control device may include a control circuit that is configured to generate a drive signal to adjust the control time of the semiconductor switch during each half-cycle of the AC mains line voltage to control the amount of power delivered to the electrical load, determine that the electrical load is an inductive load, and set the maximum current limit of the closed-loop gate drive circuit to the second limit based on the electrical load being an inductive load.

[0014] The control circuit may be configured to determine that the electrical load is an inductive load based on a magnitude of the load current conducted through the semiconductor switch being out of phase with the AC mains line voltage. The load control device may include a zero-cross detection signal that is configured to generates a zero-cross signal that indicates zero crossing points of the AC mains line voltage, and a feedback resistor that is coupled in series with the semiconductor switch and configured to generate a feedback signal. The control circuit may be configured to determine the zero-crossing points of the AC mains line voltage based on the zero-cross signal generated by the zero-cross detection signal, and determine the zero-crossing points of the load current conducted through the semiconductor switch based on the feedback signal generated by the feedback resistor. The control device may be configured to determine that the load current conducted through the semiconductor switch is out of phase with the AC mains line voltage when zero-crossings of the load current conducted through the semiconductor switch do not align with zero-crossings of the AC mains line voltage.

[0015] The control circuit may be configured to determine to operate in the forward phase-control technique based on the electrical load being an inductive load. The control circuit may be configured to detect whether to operate accordingly to a reverse phase-control technique or a forward phase-control technique based on the electrical load and, when operating in the reverse phase-control technique, the control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to the first limit. For example, when operating in the forward phase-control technique and based on the electrical load not being an inductive load, the control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to the first limit. When operating in the forward phase-control technique and based on the electrical load not being an inductive load, the control circuit may be configured to set the maximum current limit of the closed-loop gate drive circuit to the first limit for a period of time following the control time during a conduction period of the half-cycle of the AC mains line voltage and set the maximum current limit to the second limit after the expiration of the period of time during the conduction period of a half-cycle of the AC mains line voltage.

[0016] Some examples may include a load control device that is configured to control power delivered from an alternating-current (AC) power source to an electrical load, where the load control device includes a controllably conductive device that is adapted to be coupled in series between the AC power source and the electrical load, where the AC power source is configured to generate an AC mains line voltage, and where the controllably conductive device comprises a semiconductor switch configured to conduct a load current through the electrical load. The load control device may include a closed-loop gate drive circuit that is coupled to the semiconductor switch and configured to render the semiconductor switch conductive or non-conductive at a control time during half-cycles of the AC mains line voltage, where the closed-loop gate drive circuit is configured to limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit. The load control device may include a control circuit that is configured to, for an initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load, set the maximum current limit of the closed-loop gate drive circuit to a first limit for a period of time following a control time of the semiconductor switch during a conduction period of each half-cycle of the AC mains line voltage of the initial plurality of half-cycles, and set the maximum current limit of the closed-loop gate drive circuit to a second limit after the expiration of the period of time during the conduction period of each of the half-cycles of the AC mains line voltage of the initial plurality of half-cycles, wherein the second limit is less than the first limit. The control circuit may be configured to, after the initial plurality of half-cycles of the AC mains line voltage, set the maximum current limit of the closed-loop gate drive circuit to the second limit at the control time and during the entire conduction period of during subsequent half-cycles of the AC mains line voltage after the initial plurality of half-cycles while controlling power to the electrical load.

[0017] Some examples may include a load control device that is configured to control power delivered from an alternating-current (AC) power source to an electrical load, where the load control device includes a controllably conductive device that is adapted to be coupled in series between the AC power source and the electrical load, where the AC power source is configured to generate an AC mains line voltage, and where the controllably conductive device comprises a semiconductor switch configured to conduct a load current through the electrical load. The load control device may also include a control circuit that is configured to generate a drive signal to adjust a control time of the semiconductor switch during each half-cycle of the AC mains line voltage to control the amount of power delivered to the electrical load. The load control device may include a closed-loop gate drive circuit coupled to the semiconductor switch, where the closed-loop gate drive circuit is configured to receive the drive signal from the control circuit, render the semiconductor switch conductive or non-conductive at the control time in response to the drive signal during half-cycles of the AC mains line voltage, and limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit. The closed-loop gate drive circuit may be configured to adjust the maximum current limit from a first limit to a second limit during a conduction period of a half-cycle of the AC mains line voltage. In some examples, the second limit is less than the first limit. In some examples, the closed-loop gate drive circuit may be configured to render the semiconductor switch conductive at the control time when operating in a forward phase-control technique, and render the semiconductor switch non-conductive at the control time when operating in a reverse phase-control technique.

[0018] A method may be performed by a load control device that is configured to control power delivered from an alternating-current (AC) power source to an electrical load. The method may include generating a drive signal to adjust a control time of a semiconductor switch of the load control device during each half-cycle of a AC mains line voltage to control an amount of power delivered to the electrical load. The method may include controlling the drive signal to render the semiconductor switch conductive at a control time during a half-cycle of the AC mains line voltage to maintain the semiconductor switch conductive for a conduction period during the half-cycle of the AC mains line voltage. The method may include setting a maximum current limit of a closed-loop gate drive circuit to a first limit for a period of time following the control time during the conduction period of the half-cycle of the AC mains line voltage, wherein the closed-loop gate drive circuit is configured to limit a magnitude of the load current conducted through the semiconductor switch to the maximum current limit. The method may include setting the maximum current limit of the closed-loop gate drive circuit to a second limit after an expiration of the period of time during the conduction period of the half-cycle of the AC mains line voltage, wherein the second limit is less than the first limit.

[0019] The method may include setting the maximum current limit to the first limit for a pulse period that starts at the control time during the half-cycle of the AC mains line voltage. The method may include setting the maximum current limit to the second limit at an end of the pulse period and during a remainer of the conduction period of the half-cycle of the AC mains line voltage. The pulse period may be approximately 500 microseconds. The method may include setting the maximum current limit to the first limit for the pulse period to allow for a capacitance of a capacitive load to charge during the half-cycle of the AC mains line voltage.

[0020] The method may include setting the maximum current limit of the closed-loop gate drive circuit to the first limit for the pulse period following the control time of the semiconductor switch during an initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load. The method may include after the initial plurality of half-cycles of the AC mains line voltage, setting the maximum current limit of the closed-loop gate drive circuit to the second limit at the control time during subsequent half-cycles of the AC mains line voltage after the initial plurality of half-cycles and while controlling power to the electrical load.

[0021] The method may include shortening a duration of the pulse period over the initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load.

[0022] The method may include setting a length of the pulse period to a second time period for a number of half-cycles after turning on the electrical load. The method may include setting the length of the pulse period to a first time period after the number of half-cycles and while controlling power to the electrical load. In some examples, the first time period is approximately 500 microseconds, the second time period is approximately 2,000 microseconds, and the number of half-cycles is approximately 120 half-cycles.

[0023] The method may include adjusting the maximum current limit across a range of values from the first limit to the second limit over a plurality of half-cycles of the AC mains line voltage. The method may include linearly reducing the maximum current limit across the range of values from the first limit to the second limit over the plurality of half-cycles of the AC mains line voltage.

[0024] When operating in a forward phase-control technique, the method may include setting the maximum current limit of the closed-loop gate drive circuit to the first limit for the period of time following the control time of the semiconductor switch during a conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load; and setting the maximum current limit of the closed-loop gate drive circuit to the second limit after an expiration of the period of time during the conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load.

[0025] When operating in a reverse phase-control technique, the method may include setting the maximum current limit of the closed-loop gate drive circuit to a third limit during the entire conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load. In some examples, the third limit is equal to the first limit.

[0026] When using a forward phase-control technique, the method may include controlling the drive signal to render the semiconductor switch non-conductive at a beginning of each half-cycle of the AC mains line voltage; and controlling the drive signal to render the semiconductor switch conductive at the control time during the each half-cycle of the AC mains line voltage and throughout a remainder of each half-cycle of the AC mains line voltage while controlling power to the electrical load. When using a reverse phase-control technique, the method may include controlling the drive signal to render the semiconductor switches conductive at the beginning of each half-cycle of the AC mains line voltage; and controlling the drive signal to render the semiconductor switch non-conductive at the control time during each half-cycle of the AC mains line voltage.

[0027] The method may include controlling the semiconductor switch to be non-conductive for a non-conduction period and conductive for a conduction period during one or more half-cycles of the AC mains line voltage to control the amount of power delivered to the electrical load.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** is a block diagram of an example load control device for controlling the amount of power delivered to an electrical load.

[0029] FIG. **2** is a block diagram of an example gate drive circuit configured to control the amount of power delivered to an electrical load.

[0030] FIG. **3** shows examples of waveforms that illustrate operation of a load control device using a forward phase-control dimming technique.

[0031] FIG. **4** shows examples of waveforms that illustrate operation of a load control device using a reverse phase-control dimming technique.

[0032] FIG. **5** shows examples of waveforms that illustrate operation of a gate drive circuit to render a semiconductor switch conductive using a forward phase-control dimming technique.

[0033] FIG. **6** shows examples of waveforms that illustrate operation of a gate drive circuit to render a semiconductor switch conductive using a reverse phase-control dimming technique.

[0034] FIG. **7** shows examples of waveforms that illustrate operation of a gate drive circuit to render a semiconductor switch conductive using a forward phase-control dimming technique and a limit signal set to a first magnitude.

[0035] FIG. **8** shows examples of waveforms that illustrate operation of a gate drive circuit to render a semiconductor switch conductive using a forward phase-control dimming technique and a limit signal adjusted between a second magnitude and a first magnitude during a conduction period of a half-cycle of an AC mains line voltage.

[0036] FIG. **9** shows examples of waveforms that illustrate another example operation of a gate drive circuit to render a semiconductor switch conductive using a forward phase-control dimming technique and a limit signal adjusted between a second magnitude and a first magnitude during a conduction period of a half-cycle of an AC mains line voltage.

[0037] FIG. **10** shows examples of waveforms that illustrate operation of a gate drive circuit to render a semiconductor switch conductive using a forward phase-control dimming technique and a limit signal linearly adjusted between a second magnitude and a first magnitude during a conduction period of a half-cycle of an AC mains line voltage.

[0038] FIG. **11**A is a flowchart of an example procedure for setting the maximum current limit of a load control device based on the type of phase-control technique configured by the load control device.

[0039] FIG. **11**B is a flowchart of an example procedure for setting the maximum current limit of a load control device based on the type of phase-control technique configured by the load control device.

[0040] FIG. **11**C is a flowchart of an example procedure for adjusting a pulse period of a maximum current limit of a load control device in response to receiving a command to turn on the electrical load.

[0041] FIG. **12** is a flowchart of an example procedure for controlling a maximum current limit of a load control device based on whether an electrical load coupled to the load control device is an inductive load.

DETAILED DESCRIPTION

[0042] FIG. **1** is a block diagram of an example load control device **100** (e.g., a dimmer switch) for controlling an amount of power delivered from an alternating-current (AC) power source **104** to an electrical load, such as a lighting load **102**. The load control device **100** may include a hot terminal H coupled to a hot side of the AC power source **104** for receiving an AC mains line voltage V.sub.AC, and a dimmed-hot terminal DH coupled to the lighting load **102**. The load control device **100** may also include a neutral terminal N that may be adapted to be coupled to a neutral side of the AC power source **104**.

[0043] The load control device **100** may comprise a load regulation circuit **110** (e.g., a dimming circuit). For example, the load regulation circuit **110** may comprise a controllably conductive device having one or more semiconductor switches, such as two field-effect transistors (FETs) Q**111**, Q**112**. The FETs Q**111**, Q**112** may be coupled in anti-series connection between the hot terminal H and the dimmed-hot terminal DH. The load regulation circuit **110** may also include a first feedback circuit, such as a first sense resistor R**113**, coupled in series with the first FET Q**111**, and a second feedback circuit, such as a second sense resistor R**114**, coupled in series with the second FET Q**112**. The junction of the first and second sense resistors R**113**, R**114** may be coupled to circuit common. In some examples, the controllably conductive device of the load regulation circuit **110** may comprise a single FET in a full-wave bridge rectifier (e.g., coupled between the hot terminal H and the dimmed-hot terminal DH) and the load regulation circuit **110** may comprise a single closed-loop gate drive circuit.

[0044] The load control device **100** may comprise a control circuit **120**, e.g., a digital control circuit, for controlling the load regulation circuit **110** to conduct a load current I.sub.LOAD through the lighting load **102** to adjust a present intensity level L.sub.PRES of the lighting load **102**. For example, the control circuit **120** may be configured to adjust the present intensity level L.sub.PRES of the lighting load **102** between a high-end intensity level L.sub.HE (e.g., approximately 100%) and a low-end intensity level L.sub.LE (e.g., approximately 0.1-5%). The control circuit **120** may include one or more of a processor (e.g., a microprocessor), a microcontroller, a programmable logic device (PLD), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any suitable controller or processing device. The control circuit **120** may be configured to control the controllably conductive device of the load regulation circuit **110** to render the FETs Q**111**, Q**112** conductive and non-conductive using a phase-control dimming technique to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) across the lighting load **102** (e.g., between the dimmed-hot terminal DH and the neutral terminal N). When the first FET Q**111** is conductive during the positive half-cycles of the AC mains line voltage V.sub.AC, the first FET Q**111** may be configured to conduct the load current I.sub.LOAD (e.g., a first portion of the load current I.sub.LOAD) from the hot terminal H through the first sense resistor R**113** to circuit common, and then through the second sense resistor R**114** and a body diode of the second FET Q**112** to the dimmed-hot terminal DH. When the second FET Q**112** is conductive during the negative half-cycles of the AC mains line voltage V.sub.AC, the second FET Q**112** may be configured to conduct the load current I.sub.LOAD (e.g., a second portion of the load current I.sub.LOAD) from the dimmed-hot terminal DH through the second sense resistor R**114** to circuit common, and then through the first sense resistor R**113** and a body diode of the first FET Q**111** to the hot terminal H.

[0045] The control circuit **120** may be configured to determine times of zero-crossing points of the AC mains line voltage V.sub.AC of the AC power source **104**. For example, the load control device **100** may comprise a zero-crossing detect circuit **130** that may be coupled to the hot terminal H and

the neutral terminal N, and generates a zero-cross signal V.sub.ZC that indicates the zero-crossing points of the AC mains line voltage V.sub.AC. The control circuit **120** may then render the FETs Q**111**, Q**112** of the load regulation circuit **110** conductive and/or non-conductive at predetermined times (e.g., at a firing time or firing angle) relative to the zero-crossing points of the AC mains line voltage V.sub.AC (e.g., as determined from the zero-cross signal V.sub.ZC) to generate the dimmed-hot voltage V.sub.DH using the phase-control dimming technique, e.g., such as, a forward phase-control dimming technique and/or a reverse phase-control dimming technique. For example, the control circuit **120** may use the forward phase-control dimming technique to control inductive loads and may use the reverse phase-control dimming technique to control capacitive loads. During each half-cycle of the dimmed-hot voltage V.sub.DH, the control circuit **120** may be configured to control the controllably conductive device of the load regulation circuit **110** (e.g., the FETs Q**111**, Q**112**) to be non-conductive for a non-conduction period and conductive for a conduction period. In addition, the control circuit **120** may be configured to control the controllably conductive device of the load regulation circuit **110** between the non-conductive state and the conductive state during a transition period (e.g., a switching period) between the non-conduction period and the condition period. Examples of dimmer switches are described in greater detail in commonly-assigned U.S. Pat. No. 8,664,881, issued Mar. 4, 2014, entitled TWO-WIRE DIMMER SWITCH FOR LOW-POWER LOADS, the entire disclosure of which is incorporated by reference herein.
[0046] The control circuit **120** may be configured to adjust a control time (e.g., a firing time or a phase angle) each half-cycle of the dimmed-hot voltage V.sub.DH generated across the lighting load **102** each half-cycle to control the amount of power delivered to the lighting load **102** and thus the present intensity level L.sub.PRES of the lighting load **102**. For example, the control circuit **120** may be configured to adjust the control time each half-cycle to adjust the present intensity level L.sub.PRES of the lighting load **102** towards a target intensity level L.sub.TRGT. The control circuit **120** may be configured to control the FETs Q**111**, Q**112** using the forward phase-control dimming technique and/or the reverse phase-control dimming technique. When using the forward phase-control dimming technique, the control circuit **120** may render one or more of the FETs Q**111**, Q**112** non-conductive (e.g., to cause the controllably conductive device of the load regulation circuit **110** to be non-conductive) at the beginning of each half-cycle of the AC mains line voltage, and then render one or more of the FETs Q**111**, Q**112** conductive (e.g., to cause the controllably conductive device of the load regulation circuit **110** to be conductive) at the control time (e.g., the firing time) during the half-cycle after which the controllably conductive device may remain conductive until the end of the half-cycle. When using the reverse phase-control dimming technique, the control circuit **120** may render one or more of the semiconductor switches Q**111**, Q**112** conductive (e.g., to cause the controllably conductive device of the load regulation circuit **110** to be conductive) at the beginning of each half-cycle of the AC mains line voltage, and then render one or more of the semiconductor switches Q**111**, Q**112** non-conductive (e.g., to cause the controllably conductive device of the load regulation circuit **110** to be non-conductive) at the control time during the half-cycle after which the controllably conductive device may remain non-conductive until the end of the half-cycle.
[0047] The load regulation circuit **110** may also include a first gate drive circuit **115** (e.g., a first closed-loop gate drive circuit) coupled to a gate of the first FET Q**111** for controlling the first FET Q**111** and a second gate drive circuit **116** (e.g., a second closed-loop gate drive circuit) coupled to a gate of the second FET Q**112** for controlling the second FET Q**112**. The control circuit **120** may generate, for example, first and second drive signal V.sub.DR1, V.sub.DR2, respectively, that may be received by the first and second gate drive circuits **115**, **116**, respectively, for controlling the FETs Q**111**, Q**112**. The drive signals V.sub.DR1, V.sub.DR2 may be provided to the first and second gate drive circuits **115**, **116** of the load regulation circuit **110**, respectively for adjusting the control time of the dimmed-hot voltage V.sub.DH generated across the lighting load **102** and/or a magnitude of the load current I.sub.LOAD conducted through the lighting load **102**, for example,

to control the present intensity level L.sub.PRES of the lighting load **102**. The control circuit **120** may adjust a duty cycle (e.g., an on time) of each of the drive signals V.sub.DR1, V.sub.DR2 (e.g., using the same duty cycle for each of the drive signals V.sub.DR1, V.sub.DR2) to adjust the present intensity level L.sub.PRES of the lighting load **102** towards the target intensity level L.sub.TRGT.

[0048] The first and the second gate drive circuits **115**, **116** may receive the respective drive signals V.sub.DR1, V.sub.DR2 from the control circuit **120** and control the respective FETs Q**111**, Q**112** in response to the drive signals V.sub.DR1, V.sub.DR2 to adjust the magnitude of the load current I.sub.LOAD conducted through the lighting load **102**. For example, the first and second gate drive circuits **115**, **116** may generate respective gate control signals V.sub.GC1, V.sub.GC2 in response to the drive signals V.sub.DR1, V.sub.DR2, and provide the respective gate control signals V.sub.GC1, V.sub.GC2 to the gates of the respective FETs Q**111**, Q**112** to render the FETs Q**111**, Q**112** conductive and non-conductive and control the present intensity level L.sub.PRES of the lighting load **102**. While not shown in FIG. **1**, in some examples, the load control device **100** may comprise a first buffer circuit coupled between the first gate drive circuit **115** and the gate of the first FETs Q**111**, and a second buffer circuit coupled between the second gate drive circuit **116** and the gate of the second FET Q**112**.

[0049] The load regulation circuit **110** may include first and second feedback circuits, such as a first sense resistor R**113** and a second sense resistor R**114**, respectively. For example, the first sense resistor R**113** (e.g., the first feedback circuit) may be configured to generate a first feedback signal V.sub.FB1, which may have a magnitude that indicates a magnitude of the first portion of the load current I.sub.LOAD that flows through the first FET Q**111** during the positive half-cycles. In addition, the second sense resistor R**114** (e.g., the second feedback circuit) may be configured to generate a second feedback signal V.sub.FB2, which may have a magnitude that indicates a magnitude of the second portion of the load current I.sub.LOAD that flows through the second FET Q**112** during the negative half-cycles. Together, the first and second feedback signals V.sub.FB1, V.sub.FB2 may indicate the magnitude of the load current I.sub.LOAD that is conducted through the lighting load **102** (e.g., the sum of the first portion and the second portion of the load current I.sub.LOAD). The first and second gate drive circuits **115**, **116** may be configured to receive the first and second feedback signals V.sub.FB1, V.sub.FB2, respectively, and to generate the first and second gate control signals V.sub.GC1, V.sub.GC2 in response to the first and second feedback signals V.sub.FB1, V.sub.FB2, respectively (e.g., using closed-loop control). Since the first and second gate drive circuits **115**, **116** generate the first and second gate control signals V.sub.GC1, V.sub.GC2 using closed-loop control, the magnitudes of the first and second gate control signals V.sub.GC1, V.sub.GC2 may indicate the magnitude of the load current I.sub.LOAD. Examples of a load control device that provides closed-loop control of a load current conducted through a lighting load are described in greater detail in U.S. Pat. No. 11,569,733, issued Jan. 31, 2023, entitled LOAD CONTROL DEVICE HAVING A CLOSED-LOOP GATE DRIVE CIRCUIT INCLUDING OVERCURRENT PROTECTION, the entire disclosure of which is hereby incorporated by reference.

[0050] The first gate drive circuit **115** may receive the first feedback signal V.sub.FB1 and may generate the first gate control signal V.sub.GC1 for controlling the first FET Q**111** in response to the first drive signal V.sub.DR1 and the first feedback signal V.sub.FB1. For example, the first gate drive circuit **115** may be configured to adjust the magnitude of the first gate control signal V.sub.GC1 in response to a magnitude of the first feedback signal V.sub.FB1 to adjust the magnitude of the load current I.sub.LOAD towards a target current indicated by a magnitude of a first target signal V.sub.TRGT1 of the first gate drive circuit **115**. The first gate drive circuit **115** may be configured generate the first gate control signal V.sub.GC1 for controlling the magnitude of the load current I.sub.LOAD conducted through the first FET Q**111** in response to the first target signal V.sub.TRGT1 and the first feedback signal V.sub.FB1 using closed-loop control (e.g., as described in greater detail below). The first gate drive circuit **115** may be configured to receive a

first limit signal V.sub.LMT1, which may also be generated by the control circuit **120**. The first gate drive circuit **115** may be configured to generate the first target signal V.sub.TRGT1 to control the first FET Q**111** during the conduction period of each half-cycle of the dimmed-hot voltage V.sub.DH in response to the first limit signal V.sub.LMT1. For example, the magnitude of the first target signal V.sub.TRGT1 of the first gate drive circuit **115** may set a first maximum current limit I.sub.LMT1 to which the first gate control circuit **115** may control the magnitude of the load current I.sub.LOAD during the conduction period of each positive half-cycle.
[0051] The first gate drive circuit **115** may be configured to generate the first target signal V.sub.TRGT1 to control the first FET Q**111** during the transition period of each half-cycle of the dimmed-hot voltage V.sub.DH in response to the first drive signal V.sub.DR1. In some examples, the first target signal V.sub.TRGT1 of the first gate drive circuit **115** may be a shaped (e.g., wave-shaped) target signal based on the first drive signal V.sub.DR1, and the first gate drive circuit **115** may generate the first gate control signal V.sub.GC1 for turning on the first FET Q**111** in response to the shaped target signal and the first feedback signal V.sub.FB1 to shade the dimmed-hot voltage V.sub.DH during the transition period (e.g., as described in greater detail below). For example, the control circuit **120** may be configured to generate the first drive signal V.sub.DR1, and the first gate drive circuit **115** may be configured to shape (e.g., start shaping) the first target signal V.sub.TRGT1 in response to the first drive signal V.sub.DR1.
[0052] Similarly, the second gate drive circuit **116** may receive the second feedback signal V.sub.FB2 and may generate the second gate control signal V.sub.GC1 for controlling the first FET Q**111** in response to the second drive signal V.sub.DR2 and the second feedback signal V.sub.FB2. For example, the second gate drive circuit **116** may be configured to adjust a magnitude of the second gate control signal V.sub.GC2 in response to a magnitude of the second feedback signal V.sub.FB2 to adjust the magnitude of the load current I.sub.LOAD towards a target current indicated by a magnitude of a second target signal V.sub.TRGT2 of the second gate drive circuit **116**. The second gate drive circuit **116** may be configured generate the second gate control signal V.sub.GC2 for controlling the magnitude of the load current I.sub.LOAD conducted through the second FET Q**112** in response to the second target signal V.sub.TRGT2 and the second feedback signal V.sub.FB2 using closed-loop control (e.g., as described in greater detail below). The second gate drive circuit **116** may be configured to receive a second limit signal V.sub.LMT2, which may also be generated by the control circuit **120**. The second gate drive circuit **116** may be configured to generate the second target signal V.sub.TRGT2 to control the second FET Q**112** during the conduction period of each half-cycle of the dimmed-hot voltage V.sub.DH in response to the second limit signal V.sub.LMT2. For example, the magnitude of the second target signal V.sub.TRGT2 of the second gate drive circuit **116** may set a second maximum current limit I.sub.LMT2 to which the second gate control circuit **116** may control the magnitude of the load current I.sub.LOAD during the conduction period of each negative half-cycle.
[0053] The second gate drive circuit **116** may be configured to generate the second target signal V.sub.TRGT2 to control the second FET Q**112** during the transition period of each half-cycle of the dimmed-hot voltage V.sub.DH in response to the second drive signal V.sub.DR2. In some examples, the second target signal V.sub.TRGT2 of the second gate drive circuit **116** may be a shaped (e.g., wave-shaped) target signal based on the second drive signal V.sub.DR2, and the second gate drive circuit **116** may generate the second gate control signal V.sub.GC2 for turning on the second FET Q**112** in response to the shaped target signal and the second feedback signal V.sub.FB2 to shade the dimmed-hot voltage V.sub.DH during the transition period (e.g., as described in greater detail below). For example, the control circuit **120** may be configured to generate the second drive signal V.sub.DR2, and the second gate drive circuit **116** may be configured to shape (e.g., start shaping) the second target signal V.sub.TRGT2 in response to the second drive signal V.sub.DR2.
[0054] The load regulation circuit **110** (e.g., the first and second gate drive circuits **115**, **116**) may

be configured to provide overcurrent protection (e.g., inherent overcurrent protection and/or current-limiting protection) for the load control device **100**. In some examples, the first and second gate drive circuits **115**, **116** may be configured to limit the magnitude of the load current I.sub.LOAD conducted through the respective FETs Q**111**, Q**112** in response to the respective gate control signal V.sub.GC1, V.sub.GC2. For example, when the first FET Q**111** is conductive during the positive half-cycles, the first gate drive circuit **115** may be configured to limit the magnitude of the first portion of the load current I.sub.LOAD conducted through the first FET Q**111** to the first maximum current limit I.sub.LMT1 as indicated by the first limit signal V.sub.LMT1. Similarly, when the second FET Q**112** is conductive during the negative half-cycles, the second gate drive circuit **116** may be configured to limit the magnitude of the second portion of the load current I.sub.LOAD conducted through the second FET Q**112** to the second maximum current limit I.sub.LMT2 as indicated by the second limit signal V.sub.LMT2. For example, the second maximum current limit I.sub.LMT2 may be equal to the first maximum current limit I.sub.LMT1.
[0055] The load regulation circuit **110** may comprise first and second overcurrent protection (OCP) circuits **117**, **118** for detecting overcurrent conditions in the FETs Q**111**, Q**112**, respectively, and controlling the FETs Q**111**, Q**112**, respectively, in response to detecting the overcurrent conditions. For example, the first overcurrent protection circuit **117** may be configured to detect the overcurrent condition, and render the first FET Q**111** non-conductive after a first trip time period T.sub.TRIP1 from when the overcurrent condition is detected. Similarly, the second overcurrent protection circuit **118** may be configured to detect the overcurrent condition, and render the second FET Q**112** non-conductive after a second trip time period T.sub.TRIP2 from when the overcurrent condition is detected. For example, the first overcurrent threshold I.sub.TH-OCP may be equal to the second overcurrent threshold I.sub.TH-OCP2, and the first trip time period T.sub.TRIP1 may be equal to the second trip time period T.sub.TRIP2. The first and second overcurrent protection circuits **117**, **118** may be configured to receive respective overcurrent protection (OCP) enable signals V.sub.OCP-EN1, V.sub.OCP-EN2, which may be generated by the control circuit **120** for enabling the overcurrent protection circuits **117**, **118** to detect the overcurrent condition and render the FETs Q**111**, Q**112** non-conductive. For example, the control circuit **120** may be configured to enable the overcurrent protection circuits **117**, **118** during the conduction periods of each half-cycle.
[0056] The first overcurrent protection circuit **117** may receive (e.g., may be responsive to) the first gate control signal V.sub.GC1 generated by the first gate drive circuit **115**, and the second overcurrent protection circuit **118** may receive (e.g., may be responsive to) the second gate control signal V.sub.GC2 generated by the second gate drive circuit **116**. Since the first and second gate drive circuits **115**, **116** generate the first and second gate control signals V.sub.GC1, V.sub.GC2 using closed-loop control and the magnitudes of the first and second gate control signals V.sub.GC1, V.sub.GC2 may indicate the magnitude of the load current I.sub.LOAD, the first and second overcurrent protection circuit **117**, **118** may be configured to determine an overcurrent condition in the load current I.sub.LOAD (e.g., in the first FET Q**111** and/or the second FET Q**112**) in response to the first and second gate control signals V.sub.GC1, V.sub.GC2, respectively. For example, the first overcurrent protection circuit **117** may be configured to detect the overcurrent condition by determining when the first gate control signal V.sub.GC1 indicates that the magnitude of the load current I.sub.LOAD is at the first maximum current limit I.sub.LMT1 during the positive half-cycles. In addition, the second overcurrent protection circuit **118** may be configured to detect the overcurrent condition by determining when the second gate control signal V.sub.GC2 indicates that the magnitude of the load current I.sub.LOAD is at the second maximum current limit I.sub.LMT2 during the negative half-cycles.
[0057] The first and second overcurrent protection circuits **117**, **118** may be configured to control the first and second gate drive circuits **115**, **116**, respectively, to render the respective FETs Q**111**, Q**112** non-conductive in response to detecting the overcurrent condition. The first and second gate

drive circuits **115**, **116** may be configured to render the respective FETs Q**111**, Q**112** non-conductive by driving the magnitude of the first and second gate control signals V.sub.GC1, V.sub.GC2, respectively, to approximately zero volts. For example, the first gate drive circuit **115** may be configured to control the magnitude of the first gate control signals V.sub.GC1 to approximately zero volts by controlling the magnitude of the first target signal V.sub.TRGT1 of the first gate drive circuit **115** to approximately zero volts. In addition, the second gate drive circuit **116** may be configured to control the magnitude of the second gate control signals V.sub.GC1 to approximately zero volts by controlling the magnitude of the second target signal V.sub.TRGT2 of the second gate drive circuit **116** to approximately zero volts. While FIG. **1** shows the load regulation circuit **110** comprising the two different overcurrent protection circuits (e.g., the first and second overcurrent protection circuits **117**, **118**), the load regulation circuit **110** may also comprise a single overcurrent protection circuit that may be responsive to both the first gate control signals V.sub.GC1 and the second gate control signals V.sub.GC2, and may be configured to control the first gate drive circuit **115** and the second gate drive circuit **116** in response to detecting an overcurrent condition.

[0058] The load control device **100** may include a user interface **132**. The user interface **132** may include one or more actuators (e.g., buttons) for receiving user inputs and/or one or more visual indicators for providing user feedback. For example, the user interface **132** may include a toggle actuator and/or an intensity adjustment actuator (e.g., such as a slider control or a pair of raise and lower buttons) for controlling the lighting load **102**. The control circuit **120** may be configured to control the load regulation circuit **110** to control the amount of power delivered to the lighting load **102** in response to actuations of the actuators of the user interface **132**. In addition, the user interface **132** may also include one or more light sources, such as light-emitting diodes (LEDs), for illuminating the visual indicators, for example, to provide a visual indication of a status and/or the present intensity level L.sub.PRES of the lighting load **102**, and/or a visual indication of a selected preset. For example, the user interface **132** may comprise a vertically-oriented linear array of visual indicators. The control circuit **120** may be coupled to the light sources for illuminating the visual indicators of the user interface **132** to provide feedback.

[0059] The load control device **100** may comprise a communication circuit **134**, e.g., such as a wired communication circuit or a wireless communication circuit. The communication circuit **134** may be communicatively coupled to the control circuit **120** for communicating (e.g., transmitting and receiving) messages (e.g., digital messages). The communication circuit **134** may be implemented as an external integrated circuit (IC) or as an internal circuit of the control circuit **120**. For example, the communication circuit **134** may include a radio-frequency (RF) transceiver coupled to an antenna for transmitting and/or receiving RF signals. The communication circuit **330** may also include, for example, an RF transmitter for transmitting RF signals, an RF receiver for receiving RF signals, or an infrared (IR) transmitter and/or receiver for transmitting and/or receiving IR signals. The control circuit **120** may be configured to receive messages that include commands (e.g., control data) for controlling the lighting load **102** via the communication circuit **134**. The control circuit **120** may be configured to transmit messages that include status information (e.g., feedback information) via the communication circuit **134**.

[0060] The load control device **100** may include a memory **136**. The memory **136** may be communicatively coupled to the control circuit **120** for the storage and/or retrieval of, for example, operational settings, such as, the present intensity level L.sub.PRES of the lighting load **102**. The memory **136** may be implemented as an external integrated circuit (IC) or as an internal circuit of the control circuit **120**. The memory **136** may comprise a computer-readable storage media or machine-readable storage media that maintains computer-executable instructions for performing one or more procedure and/or functions as described herein. For example, the memory **136** may comprise computer-executable instructions or machine-readable instructions that when executed by the control circuit configure the control circuit to provide one or more portions of the procedures described herein. The control circuit **120** may access the instructions from memory **136** for being executed to cause the control circuit **120** to operate as described herein, or to operate one or more other devices as described herein. The memory **136** may comprise computer-executable instructions for executing configuration software. For example, the operational characteristics stored in the memory **136** may be configured during a configuration procedure of the load control device **100**.

[0061] The load control device **100** may include a power supply **138**. The power supply **140** may receive the AC mains line voltage V.sub.AC and may generate one or more direct-current (DC) voltages for powering the circuitry of the load control device **100**. For example, the power supply **140** may be configured to generate a first DC supply voltage, such as a FET supply voltage V.sub.FET-SUP (e.g., approximately 12V) for powering the gate drive circuits **115**, **116** to drive the respective FETs Q**111**, Q**112**. In addition, the power supply **140** may be configured to generate a second DC voltage, such as a supply voltage V.sub.CC for powering the control circuit **120** and/or other low-voltage circuits of the load control device **100**.

[0062] FIG. **2** is a block diagram of an example load regulation circuit **210** of a load control device **200**, such as the load control device **100** shown in FIG. **1**. The load regulation circuit **210** may comprise a field-effect transistor (FET) Q**212**, which may be coupled in series with a lighting load (e.g., the lighting load **102**) for controlling a load current I.sub.LOAD conducted through the lighting load and thus a present intensity level L.sub.PRES of the lighting load. The load regulation circuit **210** shown in FIG. **2** may be an example of half of the load regulation circuit **110** shown in FIG. **1**. For example, the FET Q**212** may be an example of the FET Q**111** and/or the FET Q**212**, and the circuitry of the load regulation circuit **210** shown in FIG. **2** may be duplicated for each of the FETs Q**111**, Q**112** to implement the load regulation circuit **110** of FIG. **1** (e.g., with the FETs coupled in anti-series connection between a hot terminal H and a dimmed-hot terminal DH). The load regulation circuit **210** may be coupled to a hot terminal (e.g., the hot terminal H) of the load control device **200** for receiving an AC mains line voltage V.sub.AC. For example, the load regulation circuit **210** may be coupled in series between the hot terminal of the load control device **200** and the lighting load (e.g., through a body diode of the additional FET in the load regulation circuit **210**). In some examples, the load regulation circuit **210** may comprise a single FET (e.g., the FET Q**212** as shown in FIG. **2**) and the load regulation circuit **210** may be coupled inside of a full-wave bridge rectifier (e.g., with the full-wave bridge rectifier coupled between the hot terminal H and the dimmed-hot terminal DH, and circuit common coupled to the dimmed-hot terminal DH).

[0063] The load control device **200** may comprise a control circuit **250**, e.g., a digital control circuit, for controlling the load regulation circuit **210** to conduct the load current I.sub.LOAD through the lighting load to adjust the present intensity level L.sub.PRES of the lighting load. For example, the control circuit **250** may be configured to adjust the present intensity level L.sub.PRES of the lighting load between a high-end intensity level L.sub.HE (e.g., approximately 100%) and a low-end intensity level L.sub.LE (e.g., approximately 0.1-5%). The control circuit **250** may include one or more of a processor (e.g., a microprocessor), a microcontroller, a programmable logic device (PLD), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any suitable controller or processing device. The control circuit **250** may be configured to control the controllably conductive device of the load regulation circuit **210** to render the FET Q**212** conductive and non-conductive using a phase-control dimming technique to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) across the lighting load. During each half-cycle of the dimmed-hot voltage V.sub.DH, the control circuit **250** may be configured to control the load regulation circuit **210**, such that the FET Q**212** may be non-conductive for a non-conduction period and conductive for a conduction period. In addition, the control circuit **250** may be configured to control the load regulation circuit **210** to control the FET Q**212** between the non-conductive state and the conductive state during a transition period (e.g., a switching period) between the non-conduction period and the condition period.

[0064] The load regulation circuit **210** may comprise a feedback circuit, such as a sense resistor R**214** (e.g., one of the resistors R**113**, R**114**) coupled in series with the FET Q**212** for conducting the load current I.sub.LOAD. In addition, the load regulation circuit **210** may comprise a gate drive circuit **220** (e.g., one of the gate drive circuits **115**, **116**) coupled to a gate of the FET Q**212** for render the FET Q**212** conductive and non-conductive. The control circuit **250** may control a drive signal V.sub.DR (e.g., control the magnitude of the drive signal V.sub.DR high or low) that may be received by the gate drive circuit **220** for controlling the FET Q**212**. The drive signal V.sub.DR1 may be provided to the gate drive circuit **220** for adjusting a control time of the dimmed-hot voltage V.sub.DH generated across the lighting load and/or a magnitude of the load current I.sub.LOAD conducted through the lighting load, for example, to control the present intensity level L.sub.PRES of the lighting load. The control circuit **250** may be configured to adjust a duty cycle (e.g., an on time) of the drive signal V.sub.DR to adjust the present intensity level L.sub.PRES of the lighting load towards a target intensity level L.sub.TRGT.
[0065] The gate drive circuit **220** may receive the drive signal V.sub.DR from the control circuit **250** and control the FET Q**212** in response to the drive signal V.sub.DR to adjust the magnitude of the load current I.sub.LOAD conducted through the lighting load. For example, the gate drive circuit **220** may generate a gate control signal V.sub.GC for controlling the FET Q**212** in response to the drive signal V.sub.DR. The load regulation circuit **210** may include a buffer circuit **216** coupled in series between the gate drive circuit **220** and the gate of the FET Q**212**. The buffer circuit **216** may receive the control signal V.sub.GC from the gate drive circuit **220** and may buffer the gate control signal V.sub.GC to generate a gate voltage V.sub.G at the gate of the FET Q**212**. The gate signal V.sub.G may control conductivity of the FET Q**212**. For example, the gate signal V.sub.G, via the gate control signal V.sub.GC, may render the FET Q**212** conductive and non-conductive. In some examples, the load regulation circuit **210** may not include the buffer circuit **216**.
[0066] The gate drive circuit **220** may be configured to receive a feedback signal V.sub.FB that may be generated by the sense resistor R**214** (e.g., the feedback circuit). The feedback signal V.sub.FB may have a magnitude that indicates the magnitude of the load current I.sub.LOAD that flows through the FET Q**212**. For example, the magnitude of the feedback signal V.sub.FB (e.g., the voltage generated across the sense resistor R**214**) may be proportional to the magnitude of the load current I.sub.LOAD. The gate drive circuit **220** may be configured to generate the gate control signal V.sub.GC in response to the feedback signal V.sub.FB (e.g., using closed-loop control). Since the gate drive circuit **212** generates the gate control signal V.sub.GC using closed-loop control, the magnitude of the gate control signal V.sub.GC may indicate the magnitude of the load current I.sub.LOAD. Further, in some examples, the control circuit **250** may receive the feedback signal V.sub.FB, and the control circuit may be configured to determine the zero-crossing points of the load current I.sub.LOAD conducted through the FET Q**212**) based on the feedback signal V.sub.FB (e.g., since the magnitude of the load current I.sub.LOAD is proportional to the magnitude of the feedback signal V.sub.FB generated by the sense resistor R**214**).
[0067] To provide closed-loop control of the magnitude of the load current I.sub.LOAD, the gate drive circuit **220** may comprise an operational amplifier circuit **222**. The operational amplifier circuit **222** may have an inverting input configured to receive the feedback signal V.sub.FB from the sense resistor R**214**. The operational amplifier circuit **222** may also have a non-inverting input configured to receive a target signal V.sub.TRGT that may indicate a target magnitude I.sub.TRGT towards which the magnitude of the load current I.sub.LOAD may be controlled. The operational amplifier circuit **222** may generate the gate control signal V.sub.GC at an output based on the target signal V.sub.TRGT and the feedback signal V.sub.FB. For example, the operational amplifier circuit **222** may adjust the gate control signal V.sub.GC to control the FET Q**212** and thus control the magnitude of the load current I.sub.LOAD towards the target magnitude I.sub.TRGT indicated by the magnitude of the target signal V.sub.TRGT. The operational amplifier circuit **222** may be

referenced to circuit common of the gate drive circuit **220** (e.g., to which the feedback signal V.sub.FB is referenced). As previously mentioned, the magnitude of the feedback signal V.sub.FB may be proportional to the magnitude of the load current I.sub.LOAD. Accordingly, the operational amplifier circuit **222** may control the magnitude of the gate control signal V.sub.GC in response to the magnitude of the load current I.sub.LOAD. Some lighting loads, such as LED light sources, are capacitive loads, which may cause changes in a load voltage (e.g., a dimmed-hot voltage) across the lighting load that are not representative of the magnitude of the load current I.sub.LOAD (e.g., proportional to the magnitude of the load current I.sub.LOAD). Since both the sense resistor R**214** and the operational amplifier circuit **222** are referenced to circuit common, the operational amplifier circuit **222** may be responsive to (e.g., only responsive to) the magnitude of the load current I.sub.LOAD and not influenced by the magnitude of the load voltage across the lighting load.

[0068] The gate drive circuit **220** may comprise a target voltage set circuit **224**, which may be configured to generate the target signal V.sub.TRGT that is received by the operational amplifier circuit **222**. The target voltage set circuit **224** of the gate drive circuit **220** may be configured to receive a limit signal V.sub.LMT, which may be generated by the control circuit **250**. The target voltage set circuit **224** may be configured to generate the target signal V.sub.TRGT to control the FET Q**212** during the conduction period of each half-cycle of the dimmed-hot voltage V.sub.DH in response to the limit signal V.sub.LMT. The target voltage set circuit **224** may be configured to set the magnitude of the target signal V.sub.TRGT based on the limit signal V.sub.LMT (e.g., a magnitude of the limit signal V.sub.LMT). For example, the magnitude of the target signal V.sub.TRGT of the first gate drive circuit **115** (e.g., based on the limit signal V.sub.LMT) may set a maximum current limit I.sub.LMT to which the gate control circuit **220** may control the magnitude of the load current I.sub.LOAD during the conduction period of each half-cycle. For example, to control the magnitude of the maximum current limit I.sub.LMT to two different respective magnitudes, the control circuit **250** may be configured to generate the limit signal V.sub.LMT at two magnitudes, for example, a first magnitude, such as a lower limit (e.g., approximately circuit common or zero volts) and a second magnitude, such as an upper limit (e.g., approximately the supply voltage V.sub.CC). For example, the control circuit **250** may be configured to adjust the magnitude of the limit signal V.sub.LMT (e.g., to approximately circuit common) to adjust the magnitude of the target signal V.sub.TRGT to a first limit-signal magnitude V.sub.MAG1 to adjust the maximum current limit I.sub.LMT to a first limit magnitude I.sub.MAG1 (e.g., approximately 8 A). In addition, the control circuit **250** may be configured to adjust the magnitude of the limit signal V.sub.LMT (e.g., to approximately the supply voltage V.sub.CC) to adjust the magnitude of the target signal V.sub.TRGT to a second limit-signal magnitude V.sub.MAG2 to adjust the maximum current limit I.sub.LMT to a second limit magnitude I.sub.MAG2 (e.g., approximately 20 A). In some examples, the control circuit **250** may be configured to adjust the magnitude of the limit signal V.sub.LMT to adjust the magnitude of the target signal V.sub.TRGT between the first limit-signal magnitude V.sub.MAG1 and the second limit-signal magnitude V.sub.MAG2 to adjust the maximum current limit I.sub.LMT between the first limit magnitude I.sub.MAG1 and the second limit magnitude I.sub.MAG2. For example, the magnitude of the maximum current limit I.sub.LMT may be proportional to the magnitude of the limit signal V.sub.LMT. The control circuit **250** may comprise, for example, a digital-to-analog converter (DAC) for adjusting the magnitude of the limit signal V.sub.LMT. In addition, the control circuit **250** may comprise, a filter circuit (e.g., a resistor-capacitor filter circuit) for generating the adjusting the magnitude of the limit signal V.sub.LMT in response to a pulse-width modulated (PWM) signal.

[0069] The gate drive circuit **220** may also comprise a turn-on circuit **226** and a turn-off circuit **228** for controlling the generation of the target signal V.sub.TRGT during the transition period between the non-conduction period and the conduction period of each half-cycle (e.g., depending on whether the control circuit **250** is using the forward phase-control dimming technique or the reverse phase-control dimming technique). When the forward phase-control dimming technique is used, the target voltage set circuit **224** may be configured to generate the target signal V.sub.TRGT during the transition period based on a turn-on signal V.sub.T-ON generated by the turn-on circuit **226**. The turn-on circuit **226** may be configured to receive the drive voltage V.sub.DR from the control circuit **250** and generate the turn-on signal V.sub.T-ON. When the forward phase-control dimming technique is used, the target voltage set circuit **224** may generate the target signal V.sub.TRGT based on the turn-on signal V.sub.T-ON during the transition period and then based on the limit signal V.sub.LMT during the conduction period.

[0070] The turn-on circuit **226** may be configured to start generating the turn-on signal V.sub.T-ON (e.g., control the magnitude of the turn-on signal V.sub.T-ON above approximately zero volts) in response to the control circuit **250** controlling the drive signal V.sub.DR to render the FET Q**212** conductive at the control time during the half-cycle (e.g., in response to the magnitude of the drive signal V.sub.DR going high at the control time during each half-cycle). For example, the target voltage set circuit **224** may be configured to control the magnitude of the target signal V.sub.TRGT to be approximately equal to the magnitude of the turn-on signal V.sub.T-ON during the transition period. In some examples, the turn-on circuit **226** may generate the turn-on signal V.sub.T-ON to cause the target signal V.sub.TRGT to have, for example, a linear or exponential shape during the transition period. In other examples, the turn-on circuit **226** may generate the target signal V.sub.TRGT to be shaped (e.g., having a magnitude that adjusts with respect to time over the transition period) based on the drive signal V.sub.DR (e.g., starting when the magnitude of the drive signal V.sub.DR goes high). During the transition period in the positive half-cycles, the gate drive circuit **220** may control the FET Q**212** in the linear region to adjust an impedance (e.g., a drain-source impedance) of the FET Q**212** based on the gate control signal V.sub.GC. The gate drive circuit **220** may control the FET Q**212** to adjust the impedance of the FET Q**212** in response to the feedback signal V.sub.FB to control the magnitude of the load current I.sub.LOAD towards the target magnitude I.sub.TRGT indicated by the target signal V.sub.TRGT, such that a shape of the load current I.sub.LOAD follows the shape of the target signal V.sub.TRGT (e.g., the shape of the turn-on signal V.sub.T-ON) during most or all of the transition period.

[0071] When the reverse phase-control dimming technique is used, the target voltage set circuit **224** may be configured to generate the target signal V.sub.TRGT during the transition period based on a turn-off signal V.sub.T-OFF generated by the turn-off circuit **228**. The turn-off circuit **228** may be configured to receive the drive voltage V.sub.DR from the control circuit **250** and generate the turn-off signal V.sub.T-OFF. In addition, the turn-off circuit **228** may be configured to receive a drain-voltage signal V.sub.DV (e.g., a second feedback signal) from a drain voltage sense circuit **218**, which may be coupled to the FET Q**212** (e.g., to a drain of the FET Q**212**). The drain voltage sense circuit **218** may detect a magnitude of a voltage developed across the FET Q**212** and may generate the drain-voltage signal V.sub.DV, such that a magnitude of the drain voltage signal V.sub.DV indicates the magnitude of the voltage across the FET Q**212**. When the reverse phase-control dimming technique is used, the target voltage set circuit **224** may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT during the conduction period, and the turn-off signal V.sub.T-ON and the drain voltage signal V.sub.DV during the transition period.

[0072] The turn-off circuit **228** may be configured to start shaping the turn-off signal V.sub.T-OFF (e.g., to begin controlling the magnitude of the turn-off signal V.sub.T-OFF towards zero volts) in response to the control circuit **250** controlling the drive signal V.sub.DR to render the FET Q**212** non-conductive at the control time during the half-cycle (e.g., in response to the magnitude of the drive signal V.sub.DR going low at the control time and further in response to the drain-voltage signal V.sub.DV). For example, the target voltage set circuit **224** may be configured to control the magnitude of the target signal V.sub.TRGT to be approximately equal to the magnitude of the turn-off signal V.sub.T-OFF during the transition period. In some examples, the turn-off circuit **228** may generate the turn-off signal V.sub.T-OFF to cause the target signal V.sub.TRGT to have, for example, a linear or exponential shape during the transition period. In other examples, the turn-off circuit **228** may generate the target signal V.sub.TRGT to be shaped (e.g., having a magnitude that adjusts with respect to time over the transition period) based on the drive signal V.sub.DR and/or the drain-voltage signal V.sub.DV. During the transition period in the negative half-cycles, the gate drive circuit **220** may control the FET Q**212** in the linear region to adjust the impedance (e.g., the drain-source impedance) of the FET Q**212** based on the gate control signal V.sub.GC. The gate drive circuit **220** may control the FET Q**212** to adjust the impedance of the FET Q**212** in response to the feedback signal V.sub.FB to control the magnitude of the load current I.sub.LOAD towards the target magnitude I.sub.TRGT indicated by the target signal V.sub.TRGT, such that the shape of the load current I.sub.LOAD follows the shape of the target signal V.sub.TRGT (e.g., the shape of the turn-off signal V.sub.T-OFF) during most or all of the transition period.

[0073] The target voltage set circuit **224** may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT from the control circuit **250**, the turn-on signal V.sub.T-ON from the turn-on circuit **226** (e.g., in response to the drive signal V.sub.DR), and/or the turn-off signal V.sub.T-OFF from the turn-off circuit **228** (e.g., in response to the drive signal V.sub.DR). For example, the target voltage set circuit **224** may act as a combining circuit and combine one or more of the limit signal V.sub.LMT, the turn-on signal V.sub.T-ON, and/or the turn-off signal V.sub.T-OFF to generate the target signal V.sub.TRGT. The gate drive circuit **220** (e.g., the operational amplifier circuit **222**) may generate the gate control signal V.sub.GC in response to the target signal V.sub.TRGT and the feedback signal V.sub.FB. During the condition period of each-half cycle, the gate drive circuit **220** may control the FET Q**212** in the saturation region based on the gate control signal V.sub.GC.

[0074] The gate drive circuit **220** of the load regulation circuit **210** may be configured to provide overcurrent protection (e.g., inherent overcurrent protection and/or current-limiting protection) for the load control device **200**. In some examples, the gate drive circuit **220** may be configured to limit the magnitude of the load current I.sub.LOAD conducted through the FET Q**212** in response to the gate control signal V.sub.GC. For example, when the FET Q**212** is conductive, the gate drive circuit **220** may be configured to limit the magnitude of the load current I.sub.LOAD conducted through the FET Q**212** to the maximum current limit I.sub.LMT as indicated by the limit signal V.sub.LMT. When the magnitude of the load current I.sub.LOAD conducted through the FET Q**212** is at the maximum current limit I.sub.LMT, the FET Q**212** may operate in the linear region.

[0075] The load regulation circuit **210** may also include an overcurrent protection (OCP) circuit **230** (e.g., the overcurrent protection circuits **117**, **118**). The overcurrent protection circuit **230** may comprise an overcurrent detection circuit **232** and an overcurrent override circuit **234**. The overcurrent detection circuit **232** may be configured to detect the overcurrent condition, and generate an overcurrent detection signal V.sub.OCD that indicates the overcurrent condition. The overcurrent detection circuit **232** of the overcurrent protection circuit **230** may receive (e.g., may be responsive to) the gate control signal V.sub.GC generated by the gate drive circuit **220**. Since the gate drive circuit **220** generates the gate control signal V.sub.GC1 using closed-loop control and the magnitude of the gate control signal V.sub.GC indicates the magnitude of the load current I.sub.LOAD, the overcurrent detection circuit **232** may be configured to detect the overcurrent condition in the load current I.sub.LOAD (e.g., in the FET Q**212**) in response to the gate control signal V.sub.GC. For example, the first gate drive circuit **115** may be configured to detect the overcurrent condition by determining when the gate control signal V.sub.GC indicates that the magnitude of the load current I.sub.LOAD is at the maximum current limit I.sub.LMT.

[0076] The overcurrent detection circuit **232** may be configured to receive an overcurrent protection (OCP) enable signal V.sub.OCP-EN from the control circuit **250** for enabling and disabling the overcurrent protection circuit **230**. The overcurrent detection circuit **232** may be configured to control the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition when the overcurrent protection circuit **230** is enabled (e.g., in response to the overcurrent protection enable signal V.sub.OCP-EN). The overcurrent detection circuit **232** may be configured to control the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition after a trip time period T.sub.TRIP from when the overcurrent condition is first detected. The control circuit **250** may be configured to enable the overcurrent protection circuit **230** during the conduction periods of each half-cycle (e.g., as shown in FIGS. **5** and **7-10**). For example, when operating using a forward phase-control technique, the control circuit **250** may generate the OCP enable signal V.sub.OCP-EN (e.g., drive the OCP enable signal V.sub.OCP-EN high toward the supply voltage V.sub.CC) after a time delay from the control time t.sub.CNTL. For instance, the control circuit **250** may generate the OCP enable signal V.sub.OCP-EN after a turn-on time period T.sub.T-ON (e.g., approximately 50 microseconds) from the control time t.sub.CNTL. The control circuit may drive the OCP enable signal V.sub.OCP-EN low (e.g., toward circuit common) at the next zero-crossing event. As such, the control circuit **250** may be configured to enable the overcurrent protection circuit **230** during the conduction periods of each half-cycle.

[0077] The overcurrent override circuit **234** may be configured to receive the overcurrent detection signal V.sub.OCD from the overcurrent detection circuit **232**, and generate an overcurrent protection signal V.sub.OCP for rendering the FET Q**212** non-conductive after the overcurrent detection signal V.sub.OCD controls the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition. The overcurrent override circuit **234** may be configured to control the gate drive circuit **220** to render the FET Q**212** non-conductive in response to the overcurrent detection signal V.sub.OCD controlling the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition. The gate drive circuit **220** may be configured to render the FET Q**212** non-conductive in response to the overcurrent protection signal V.sub.OCP by driving the magnitude of the gate control signal V.sub.GC to approximately zero volts. For example, the target voltage set circuit **224** may be configured to receive the overcurrent protection signal V.sub.OCP and to control the magnitude of the target signal V.sub.TRGT of the first gate drive circuit **115** to approximately zero volts to control the magnitude of the gate control signal V.sub.GC to approximately zero volts.

[0078] The overcurrent override circuit **234** may be configured to render the FET Q**212** non-conductive after the trip time period T.sub.TRIP from when the overcurrent detection circuit **232** first detects the overcurrent condition. The overcurrent override circuit **234** may be configured to latch the FET Q**212** in the non-conductive state for the remainder of the present half-cycle in response to detecting the overcurrent condition. The overcurrent override circuit **234** may be configured to receive an overcurrent protection reset signal V.sub.OCP-RST, which may be generated by the control circuit **250**. The overcurrent override circuit **234** may be configured, in response to the overcurrent protection reset signal V.sub.OCP-RST, to unlatch the FET Q**212** from the non-conductive state to allow the FET Q**212** to be rendered conductive again in response to the gate control signal V.sub.GC. For example, the control circuit **250** may be configured to control the overcurrent protection reset signal V.sub.OCP-RST to unlatch the FET Q**212** from the non-conductive state at the end of each half-cycle.

[0079] The control circuit **250** may be configured to receive the overcurrent detection signal V.sub.OCD from the overcurrent detection circuit **232**, and to provide overcurrent protection (e.g., redundant overcurrent protection) for the load control device **200** in response to the overcurrent detection signal V.sub.OCD. For example, the control circuit **250** may be configured to control the drive signal V.sub.DR to render the FETs Q**212** non-conductive when the overcurrent detection signal V.sub.OCD indicates an overcurrent condition. If the overcurrent condition persists repeatedly (e.g., for a number of half-cycles, such as 10 half-cycles in a row), the control circuit **250** may be configured to render the FET Q**212** to turn off the lighting load for a period of time (e.g., a few seconds), and then attempt to turn the lighting load back on.

[0080] When using the forward phase-control dimming technique, the control circuit **250** may control the drive signal V.sub.DR to render the FET Q**212** non-conductive (e.g., control the magnitude of the drive signal V.sub.DR low towards circuit common) at the beginning of the half-

cycle. In addition, the control circuit **250** may control the overcurrent protection enable signal V.sub.OCP-EN to disable the overcurrent protection circuit **230** during the non-conduction period of the half-cycle. At the control time during the half-cycle, the control circuit **250** may be configured to control the drive signal V.sub.DR to render the FET Q**212** conductive (e.g., control the magnitude of the drive signal V.sub.DC high towards the supply voltage V.sub.CC) to generate an edge (e.g., a rising edge) in the drive signal V.sub.DR. In response to the edge in the drive signal V.sub.DR, the target voltage set circuit **224** may set the target voltage V.sub.TRGT to be equal to the turn-on signal V.sub.T-ON during the transition period of the half-cycle. As noted above, the turn-on signal V.sub.T-ON may be shaped. At the beginning of the conduction period (e.g., at the end of the transition period), the control circuit **250** may be configured to control the overcurrent protection enable signal V.sub.OCP-EN to enable the overcurrent protection circuit **230** and control the limit signal V.sub.LMT to set the magnitude of the maximum current limit I.sub.LMT to which the gate control circuit **220** may control the magnitude of the load current I.sub.LOAD during the conduction period of the half-cycle.
[0081] When using the reverse phase-control dimming technique, the control circuit **250** may control the drive signal V.sub.DR to render the FET Q**212** conductive (e.g., control the magnitude of the drive signal V.sub.DR high towards the supply voltage V.sub.CC) at the beginning of the half-cycle. In some examples, the control circuit **250** may already be driving the magnitude of the drive signal V.sub.DR high towards the supply voltage V.sub.CC at the beginning of the half-cycle, so the control circuit **250** may simply maintain the magnitude of the drive signal V.sub.DR to render the FET Q**212** conductive at the beginning of the half-cycle. In addition, the control circuit **250** may control the overcurrent protection enable signal V.sub.OCP-EN to enable the overcurrent protection circuit **230** and control the limit signal V.sub.LMT to set the magnitude of the maximum current limit I.sub.LMT to which the gate control circuit **220** may control the magnitude of the load current I.sub.LOAD during the conduction period of the half-cycle. At the control time during the half-cycle, the control circuit **250** may be configured to control the drive signal V.sub.DR to render the FET Q**212** non-conductive (e.g., control the magnitude of the drive signal V.sub.DC low towards circuit common) to generate an edge (e.g., a falling edge) in the drive signal V.sub.DR. At the end of the conduction period (e.g., at the beginning of the transition period), the control circuit **250** may be configured to control the overcurrent protection enable signal V.sub.OCP-EN to disable the overcurrent protection circuit **230**.
[0082] In response to the edge of the drive signal V.sub.DR, the gate drive circuit **220** may be configured to decrease (e.g., start decreasing) the magnitude of the target voltage V.sub.TRGT with respect to time. For example, the turn-off circuit **228** may set the magnitude of the turn-off signal V.sub.T-OFF to a predetermined value, e.g., approximately circuit common, to set the magnitude of the maximum current limit I.sub.LMT at the first limit magnitude I.sub.MAGI (e.g., approximately 8 A). Thereafter, the turn-off circuit **228** may gradually decrease the magnitude of the turn-off signal V.sub.T-OFF until the drain voltage signal V.sub.DV from the drain voltage sense circuit **218** indicates that the voltage across the FET Q**212** is starting to rise (e.g., has exceeding a drain voltage threshold V.sub.TH indicating that the FET is starting to become non-conductive). Once the magnitude of the voltage developed across the FET Q**212** (e.g., as indicated by the drain voltage signal V.sub.DV) exceeds the drain voltage threshold V.sub.TH, the turn-off circuit **228** may begin to shape the turn-off signal V.sub.T-OFF from the present magnitude (e.g., the magnitude at the time the drain voltage signal V.sub.DV indicates that the magnitude of the voltage across the FET Q**212** has exceeded the drain voltage threshold V.sub.TH) to zero volts. The target voltage set circuit **224** may set the target voltage V.sub.TRGT to be equal to the turn-off signal V.sub.T-OFF during the rest of the transition period of the half-cycle.
[0083] The load control device **200** may be configured to use the reverse phase-control dimming technique to control a capacitive load. A capacitive load may be characterized by a large capacitance that needs to be charged when turning on the capacitive load and/or during normal

operation as the capacitance discharges. The capacitance of a capacitive load may need to draw a rather large charging current through the load control device **200** to charge the capacitance when first turning on the capacitive load and/or to recharge the capacitance during each half-cycle. When using the reverse phase-control dimming technique, the control circuit **250** may be configured to control the limit signal V.sub.LMT to adjust the magnitude of the target signal V.sub.TRGT to the second limit-signal magnitude V.sub.MAG2 (e.g., for the length of the conduction period) to set the magnitude of the maximum current limit I.sub.LMT at a larger magnitude, such as the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A), to allow the FET Q**212** to appropriately conduct the charging current of the capacitive load. Since the charging current may only last for a short amount of time, the conduction of the charging current may not damage the FET Q**212** and the overcurrent protection circuit **230** may not need to render the FET Q**212** non-conductive to protect the FET Q**212**.
[0084] In addition, the load control device **200** may be configured to use the forward phase-control dimming technique to control an inductive load. Some inductive loads may saturate when an asymmetrical current having a direct-current (DC) component is conducted through the inductive load. For example, an inductive load may be susceptible to overheating and/or damage when the inductive load conducts an asymmetrical current having a large peak magnitude above a saturation threshold in at least one of the half-cycles. In some instances, the saturation threshold may be less than the maximum current limit I.sub.LMT of the over-current protection circuit **230**. Accordingly, when using the forward phase-control dimming technique, the control circuit **250** may be configured to control the limit signal V.sub.LMT to adjust the magnitude of the target signal V.sub.TRGT to the first limit-signal magnitude V.sub.MAG1 to set the magnitude of the maximum current limit I.sub.LMT at a smaller magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A), to allow the overcurrent protection circuit **230** to appropriately protect the inductive loads from large load currents. The first limit magnitude I.sub.MAG1 may be smaller than second limit magnitude I.sub.MAG2, which for example, may prevent an inductive load from saturating and/or overheating when using the forward phase-control technique.
[0085] However, in some examples, the load control device **200** may be configured to use the forward phase-control dimming technique to control a capacitive load. When the FET Q**212** is rendered conductive each half-cycle at the control time (e.g., the firing time), the capacitive load may be configured to conduct a large, but short spike of charging current to charge the capacitance of the capacitive load. If the capacitive load is unable to conduct the large spike of charging current, the capacitive load may not function properly during normal operation (e.g., the capacitive load will not be able to establish an internal power supply or bus voltage to run its electrical components, and/or the FET Q**212** may operate in the linear region and block the charging current). However, the magnitude of the spike of charging current may exceed the maximum current limit I.sub.LMT, which may be set at the first limit magnitude I.sub.MAG1 when using the forward phase-control dimming technique (e.g., to accommodate inductive loads). As a result, the capacitive load may not be able to appropriately draw the charging current through the FET Q**212**, and the capacitance of the capacitive load may take longer to charge, which could cause flickering and/or unwanted adjustments in the present intensity level L.sub.PRES of the lighting load.
[0086] Accordingly, in some examples, when using the forward phase-control dimming technique, the control circuit **250** may be configured to increase the magnitude of the maximum current limit I.sub.LMT for a brief period of time following (e.g., immediately following) the control time (e.g., the firing time) before adjusting the maximum current limit I.sub.LMT to a smaller magnitude during the rest of the conduction period of the half-cycle. For example, the control circuit **250** may be configured to adjust the maximum current limit I.sub.LMT to the second limit magnitude I.sub.MAG2 for a pulse period T.sub.PULSE that starts at the charging time and then adjust the maximum current limit I.sub.LMT to the first limit magnitude I.sub.MAG1 during the rest of the conduction period of the half-cycle. For example, the control circuit **250** may set a length of the

pulse period T.sub.PULSE to a first time period T.sub.P1 (e.g., approximately 500 μsec) to allow the capacitance of the capacitive load to appropriately charge each half-cycle.

[0087] In some examples, the load control device **200** may be configured to use the forward phase-control dimming technique to control an incandescent lamp (e.g., a resistive load). Incandescent lamps may have a low resistance when not energized (e.g., off) and thus cold. The resistance of the incandescent lamp may increase after being energized (e.g., and thus warm) for a period of time. Because of the low resistance when cold, an incandescent lamp may draw more current (e.g., an inrush current) when first turned on than after the incandescent lamp has been energized for the period of time. The magnitude of the inrush current conducted by the incandescent lamp when first turn on may exceed the first limit magnitude I.sub.MAG1 of the maximum current limit I.sub.LMT, and as such, the incandescent load may not be able to appropriately conduct the inrush current if the maximum current limit I.sub.LMT is set to first limit magnitude I.sub.MAG1. This may cause the incandescent lamp to take longer to warm up, which may cause the present intensity level L.sub.PRES of the incandescent lamp to take longer to get to the appropriate level-even with the increased magnitude of the maximum current limit I.sub.LMT (e.g., the second limit magnitude I.sub.MAG2) during the pulse period T.sub.PULSE (e.g., as set at the first period length T.sub.P1). Accordingly, when turning on the lighting load, the control circuit **250** may be configured to increase (e.g., temporarily increase) the length of the pulse period T.sub.PULSE for a period of time to allow the incandescent lamp to conduct the inrush current. For example, when turning on the lighting load, the control circuit **250** may be configured to increase the length of the pulse period T.sub.PULSE to a second time period T.sub.P2 (e.g., approximately 2,000 μsec) for a number of half-cycles (e.g., approximately 120 half-cycles) before reverting back to the first time period T.sub.P1.

[0088] In some examples, when using the forward phase-control dimming technique, the control circuit **250** may be configured to increase the magnitude of the maximum current limit I.sub.LMT to a constant magnitude for a brief period of time following (e.g., immediately following) the control time (e.g., the firing time) before reducing the magnitude of the maximum current limit I.sub.LMT with respect to time and then maintaining the maximum current limit I.sub.LMT to a smaller magnitude during the rest of the conduction period of the half-cycle. For example, the control circuit **250** may be configured to adjust the maximum current limit I.sub.LMT to the second limit magnitude I.sub.MAG2 for the first time period T.sub.P1 and then reducing (e.g., linearly reducing) the magnitude of the maximum current limit I.sub.LMT from the second limit magnitude I.sub.MAG2 to the first limit magnitude I.sub.MAG1 across an adjustment period T.sub.ADJ before maintaining the maximum current limit I.sub.LMT to the first limit magnitude I.sub.MAG1 for the remainder of the half-cycle.

[0089] Further, in some examples, when using the forward phase-control dimming technique, the control circuit **250** may be configured to increase the magnitude of the maximum current limit I.sub.LMT over time during a ramp-up period T.sub.R-UP until the maximum current limit I.sub.LMT reaches the second limit magnitude I.sub.MAG2 following (e.g., immediately following) the control time (e.g., the firing time), maintain the maximum current limit I.sub.LMT at the second limit magnitude I.sub.MAG2 for a period of time (e.g., the pulse period T.sub.PULSE), and reduce the magnitude of the maximum current limit I.sub.LMT with respect to time over a ramp-down period T.sub.R-DOWN until the maximum current limit I.sub.LMT reaches the second limit magnitude I.sub.MAG1 (e.g., and then maintain the maximum current limit I.sub.LMT at the second limit magnitude I.sub.MAG1 for the rest of the conduction period of the half-cycle). The control circuit **250** may adjust the maximum current limit I.sub.LMT over the ramp-up and/or ramp-down periods using a linear adjustment, a step-wise adjustment, or other shapes.

[0090] In addition, the control circuit **250** may be configured to determine whether the load control device **200** is connected to an inductive load, and adjust the magnitude of the maximum current limit I.sub.LMT when the load control device **200** is connected to an inductive load. For example, the control circuit **250** may be configured to determine that the load control device **200** is connected to an inductive load if the magnitude of the load current I.sub.LOAD conducted through the FET Q**212** (e.g., as indicated by the feedback signal V.sub.FB) is out of phase with the AC mains line voltage V.sub.AC (e.g., if zero-crossings of the load current I.sub.LOAD as indicated by the feedback signal V.sub.FB do not line up with zero-crossings of the AC mains line voltage V.sub.AC as indicated by the zero-cross signal V.sub.ZC). When the control circuit **250** determines that the load control device **200** is connected to an inductive load, the control circuit **250** may control the limit signal V.sub.LMT to set the magnitude of the maximum current limit I.sub.LMT at the first limit magnitude I.sub.MAG1 while using the forward phase-control dimming technique. In some example, when the control circuit **250** determines that the load control device **200** is connected to an inductive load, the control circuit **250** may control the limit signal V.sub.LMT to set the magnitude of the maximum current limit I.sub.LMT at the second limit magnitude I.sub.MAG2 for the pulse period T.sub.PULSE and then set the magnitude of the maximum current limit I.sub.LMT to the first limit magnitude I.sub.MAG1 during the rest of the conduction period of the half-cycle while using the forward phase-control dimming technique.

[0091] FIG. **3** shows examples of waveforms that illustrate an operation of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) using a forward phase-control dimming technique. As described herein, the load control device may receive an alternating-current (AC) main line voltage from an AC power source and may generate a dimmed-hot voltage V.sub.DH at a dimmed-hot terminal of the load control device for controlling a lighting load. Using the forward phase-control dimming technique as shown in FIG. **3**, a control circuit (e.g., the control circuit **120** and/or the control circuit **250**) may render a controllably conductive device non-conductive at the beginning of each half-cycle during a non-conduction period T.sub.NC, render the controllably conductive device conductive at a control time (e.g., a firing time) during the half-cycle, and maintain the controllably conductive device conductive during a conduction period T.sub.CON until the end of the half-cycle. The control circuit may control the controllably conductive device between the non-conductive state and the conductive state during a transition time T.sub.TRAN during each half-cycle. For example, the controllably conductive device may comprise two semiconductor switches, such as two FETs in anti-series connection (e.g., as shown in FIG. **1**). Further, it should be appreciated that the slope of the dimmed-hot voltage V.sub.DH when transitioning from and to zero volts as shown in FIGS. **3-10** is exaggerated for illustrated purposes. The control circuit may generate one or more drive signals for controlling the conductivity of the controllably conductive device. For example, the control circuit may generate a first drive signal V.sub.DR1 to render a first semiconductor switch (e.g., the first FET Q**111** of the load regulation circuit **110**) conductive during the positive half-cycles, and may generate a second drive signal V.sub.DR2 to render a second semiconductor switch (e.g., the second FET Q**112** of the load regulation circuit **110**) conductive during the negative half-cycles. For example, the first and second drive signals V.sub.DR1, V.sub.DR2 may be pulse-width modulated signals.

[0092] The load control device may generate target signals (e.g., a first target signal V.sub.TRGT1 and a second target signal V.sub.TRGT2) for controlling a load current conducted through the lighting load. For example, as described herein, a first gate drive circuit of the load control device (e.g., the first gate drive circuit **115** and/or the gate drive circuit **220**) may receive the first drive signal V.sub.DR1, which may be driven high at a control time t.sub.CTNL1 (e.g., a firing time) during each of the positive half-cycles of the AC main line voltage. The first gate drive circuit may start to shape the first target signal V.sub.TRGT1 in response to the first drive signal V.sub.DR1, and may shape the first target signal V.sub.TRGT1 during a turn-on time period (e.g., during the transition period T.sub.TRAN). The first gate drive circuit may use the first target signal V.sub.TRGT1 to render the first semiconductor switch conductive and maintain the first semiconductor switch conductive during the remainder of the positive half-cycle (e.g., during the conduction period T.sub.CON). Similarly, a second gate drive circuit of the load control device (e.g., the second gate drive circuit **116** and/or the gate drive circuit **220**) may receive the second drive signal V.sub.DR2, which may be driven high at a control time t.sub.CNTL2 (e.g., a firing time) during each of the negative half-cycles of the AC main line voltage. The second gate drive circuit may shape the second target signal V.sub.TRGT2 in response to the second drive signal V.sub.DR2, and may shape the second target signal V.sub.TRGT2 during a turn-on time period (e.g., during the transition time T.sub.TRAN). The second gate drive circuit may use the second target signal V.sub.TRGT2 to render the second semiconductor switch conductive and maintain the second semiconductor switch conductive during the remainder of the negative half-cycle.

[0093] At the end of each of the negative half-cycles of the AC main line voltage (e.g., near a zero-crossing of the AC main line voltage), the first drive signal V.sub.DR1 may be driven low by the control circuit, and the first gate drive circuit may shape the first target signal V.sub.TRGT1 during a turn-off time period. Similarly, at the end of each of the positive half-cycles of the AC main line voltage (e.g., near a zero-crossing of the AC main line voltage), the second drive signal V.sub.DR2 may be driven low by the control circuit, and the second gate drive circuit may shape the second target signal V.sub.TRGT2 during a turn-off time period. Accordingly, the first and second target signals V.sub.TRGT1, V.sub.TRGT2 may be shaped on their rising and falling edges. Alternatively, the first and second target signals V.sub.TRGT1, V.sub.TRGT2 may only be shaped on their rising edges and not their falling edges when using the forward phase-control dimming technique.

[0094] FIG. **4** shows examples of waveforms that illustrate an operation of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) using a reverse phase-control dimming technique. Similar to the operation of the load control device with a forward phase-control dimming technique described herein, the load control device may receive an AC mains line voltage from an AC power source and may generate a dimmed-hot voltage V.sub.DH at a dimmed-hot terminal of the load control device for controlling a lighting load. Using the reverse phase-control dimming technique as shown in FIG. **4**, a control circuit may render a controllably conductive device conductive at the beginning of each half-cycle during a conduction period T.sub.CON, render the controllably conductive device non-conductive at a control time (e.g., a firing time) during the half-cycle, and maintain the controllably conductive device conductive during a non-conduction period T.sub.NC until the end of the half-cycle. The control circuit may control the controllably conductive device between the non-conductive state and the conductive state during a transition time T.sub.TRAN during each half-cycle. For example, the controllably conductive device may comprise two semiconductor switches, such as two FETs in anti-series connection. The control circuit may generate one or more drive signals for controlling the conductivity of the controllably conductive device. For example, the control circuit may generate a first drive signal V.sub.DR1 to render a first semiconductor switch (e.g., the first FET Q**111** of the load regulation circuit **110**) conductive during the positive half-cycles. The control circuit may generate a second drive signal V.sub.DR2 to render a second semiconductor switch (e.g., the second FET Q**112** of the load regulation circuit **110**) conductive during the negative half-cycles. For example, the first and second drive signals V.sub.DR1, V.sub.DR2 may be pulse-width modulated signals.

[0095] The load control device may generate target signals (e.g., a first target signal V.sub.TRGT1 and a second target signal V.sub.TRGT2) for controlling a load current conducted through the lighting load. For example, as described herein, a first gate drive circuit of the load control device (e.g., the first gate drive circuit **115** and/or the gate drive circuit **220**) may receive the first drive signal V.sub.DR1, which may be high at the beginning of each of the positive half-cycles, such that the controllably conductive device is rendered conductive at the beginning of each of the positive half-cycles. The first drive signal V.sub.DR-1a may be driven low at a control time t.sub.CNTL1 (e.g., a firing time) during each of the positive half-cycles of the AC mains line voltage. The first gate drive circuit may begin to shape the first target signal V.sub.TRGT-1a in response to the first drive signal V.sub.DR1, and may shape the first target signal V.sub.TRGT1 during a turn-off time period (e.g., during the transition period T.sub.TRAN). The first gate drive circuit may use the first target signal V.sub.TRGT1 to render the first semiconductor switch non-conductive and maintain the first semiconductor switch conductive during the remainder of the positive half-cycle (e.g., during the non-conduction period T.sub.NC). Similarly, a second gate drive circuit of the load control device (e.g., the second gate drive circuit **116** and/or the gate drive circuit **220**) may receive the second drive signal V.sub.DR2, which may be high at the beginning of each of the negative half-cycles, such that the controllably conductive device is rendered conductive at the beginning of each of the negative half-cycles. The second drive signal V.sub.DR2 may be driven low at a control time t.sub.CNTL2 (e.g., a firing time) during each of the negative half-cycles of the AC mains line voltage. The load control device may begin to shape the second target signal V.sub.TRGT2 in response to the second drive signal V.sub.DR2, and may shape the second target signal V.sub.TRGT2 during a turn-off time period (e.g., during the transition period T.sub.TRAN). The second gate drive circuit may use the second target signal V.sub.TRGT2 to render the second semiconductor switch conductive during the remainder of the negative half-cycle (e.g., during the non-conduction period T.sub.NC)

[0096] At the end of each of the positive half-cycles of the AC mains line voltage (e.g., near a zero-crossing of the AC mains line voltage), the first drive signal V.sub.DR1 may be driven high by the control circuit, and the first gate drive circuit may shape the first target signal V.sub.TRGT1 during a turn-on time period. Similarly, at the end of each of the negative half-cycles of the AC mains line voltage (e.g., near a zero-crossing of the AC mains line voltage), the second drive signal V.sub.DR2 by be drive high by the control circuit, and the second gate drive circuit may shape the second target signal V.sub.TRGT2 during a turn-on time period. Accordingly, the first and second target signals V.sub.TRGT1, V.sub.TRGT2 may be shaped on their rising and falling edges. Alternatively, the first and second target signals V.sub.TRGT1, V.sub.TRGT2 may only be shaped on their falling edges and not their rising edges when using the reverse phase-control dimming technique.

[0097] FIG. **5** shows examples of waveforms that illustrate operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) conductive using a forward phase-control dimming technique. In the waveforms of FIG. **5**, the lighting load may not be experiencing an overcurrent event (e.g., the lighting load is shorted) or an inrush current event (e.g., conducting an inrush current to the lighting load). As described herein, a load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may use a phase control dimming technique to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device for controlling a lighting load. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **5**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half-cycles may be the same as the operation of the first gate drive circuit in the positive half-cycles as shown in FIG. **5**.

[0098] The gate drive circuit of the load control device may receive a drive signal V.sub.DR from a control circuit (e.g., the control circuit **120** and/or the control circuit **250**). The control circuit may control a magnitude of the drive signal V.sub.DR high (e.g., towards the DC supply voltage V.sub.CC) at a control time t.sub.CNTL (e.g., a firing time). The gate drive circuit may generate a target signal V.sub.TRGT in response to the drive signal V.sub.DR from the control circuit. The gate drive circuit may begin to shape the target signal V.sub.TRGT in response to the drive signal V.sub.DR being driven high. The gate drive circuit may shape the target signal V.sub.TRGT over a turn-on time period T.sub.T-ON (e.g., approximately 50 microseconds). In some examples, the transition period T.sub.TRAN may be equal to the turn-on time period T.sub.T-ON, for example, when using forward phase-control. The gate drive circuit may generate the target signal V.sub.TRGT using a turn-on signal V.sub.T-ON (not shown) generated by a turn-on circuit (e.g., the turn-on circuit **226**). For example, the turn-on signal V.sub.T-ON may have the same shape as the target voltage V.sub.TRGT during the turn-on period T.sub.T-ON as shown in FIG. **5**. The gate drive circuit may shape the turn-on signal V.sub.T-ON in order to shape the target signal V.sub.TRGT. In some examples, the turn-on signal V.sub.T-ON may be defined by an “S” shape over the turn-on time period T.sub.T-ON. The gate drive circuit may control the magnitude of the turn-on signal V.sub.T-ON to the first limit-signal magnitude V.sub.MAG1 at the end of the turn-on time period T.sub.T-ON. The gate drive circuit may set the target signal V.sub.TRGT to be equal to the magnitude of the turn-on signal V.sub.T-ON during the turn-on time period T.sub.T-ON.

[0099] After the turn-on time period T.sub.T-ON, the gate drive circuit may be configured to set the target signal V.sub.TRGT based on the turn-on signal V.sub.T-ON and a limit signal V.sub.LMT. The magnitude of the limit signal V.sub.LMT may range between a second magnitude (e.g., approximately the supply voltage V.sub.CC) and a first magnitude (e.g., approximately circuit common or zero volts). As noted herein, the control circuit may set the magnitude of the limit signal V.sub.LMT at the second magnitude or the first magnitude (e.g., or one or more values between the upper and first magnitudes) to control the magnitude of the maximum current limit I.sub.LMT of the load control device. In the waveforms of FIG. **5**, the magnitude of the limit signal V.sub.LMT is set to the second magnitude at the end of the turn-on time period T.sub.T-ON, and as such, the magnitude of the target signal V.sub.TRGT may be equal to the second limit-signal magnitude V.sub.MAG2 at the end of the turn-on time period T.sub.T-ON.

[0100] Further, after the turn-on time period T.sub.T-ON, the control circuit may control the overcurrent protection enable signal V.sub.OCP-EN to enable the overcurrent protection circuit. In some examples, the control circuit may control the overcurrent protection enable signal V.sub.OCP-EN to enable the overcurrent protection circuit after a delay period (e.g., approximately 50 microseconds) after the end of the turn-on time period T.sub.T-ON. As noted herein, the overcurrent detection circuit may be configured to control the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition after a trip time period T.sub.TRIP from when the overcurrent condition is first detected. The control circuit may be configured to enable the overcurrent protection circuit during the conduction periods of each half-cycle, for example, as shown in FIG. **5**. For example, when operating using the forward phase-control technique, the control circuit may drive the magnitude of the OCP enable signal V.sub.OCP-EN high (e.g., toward the supply voltage V.sub.CC) after the turn-on time period T.sub.T-ON. The control circuit may drive the magnitude of the OCP enable signal V.sub.OCP-EN low (e.g., toward circuit common) at and/or around the next zero-crossing event. As such, the control circuit may be configured to enable the overcurrent protection circuit during the conduction period of each half-cycle.

[0101] The gate drive circuit may generate a gate control signal V.sub.GC based on the target signal V.sub.TRGT and a feedback signal V.sub.FB (not shown) that indicates a magnitude of a load current I.sub.LOAD conducted through the semiconductor switch. The gate drive circuit may adjust a magnitude of the gate control signal V.sub.GC in response to the magnitude of the feedback signal V.sub.FB. For example, the gate drive circuit may adjust the magnitude of the gate control signal V.sub.GC in response to a magnitude of the feedback signal V.sub.FB to adjust the magnitude of the load current I.sub.LOAD toward a target current, where for example, the target current is indicated by a magnitude of the target signal V.sub.TRGT.

[0102] When the magnitude of the target signal V.sub.TRGT is equal to the second limit-signal magnitude V.sub.MAG2 at the end of the turn-on time period T.sub.T-ON, the gate drive circuit may control the magnitude of the gate control signal V.sub.GC to attempt to control the magnitude of the load current I.sub.LOAD to the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A). However, the lighting load may not require the second limit magnitude I.sub.MAG2, and the magnitude of the gate control signal V.sub.GC may deviate from the magnitude of the target signal V.sub.TRGT. When the magnitude of the target signal V.sub.TRGT is equal the second limit magnitude I.sub.MAG2 after the turn-on time period T.sub.T-ON, the gate drive circuit may control the magnitude of the gate control signal V.sub.GC to attempt to control the magnitude of the load current I.sub.LOAD to a maximum current level I.sub.MAX (e.g., approximately 20 A). However, if the lighting load is not experiencing an overcurrent event (e.g., the lighting load is shorted) or an inrush current event (e.g., conducting an inrush current to the lighting load, such as an incandescent load), and an input capacitor of the lighting load does not create a surge of current, the magnitude of the gate control signal V.sub.GC may not be driven up to the second limit magnitude I.sub.MAG2, as shown in FIG. **5**.

[0103] Since the gate drive circuit is using closed-loop control to adjust the magnitude of the gate control signal V.sub.GC based on the feedback signal V.sub.FB, the magnitude of the gate control signal V.sub.GC may indicate (e.g., be proportional to) the present magnitude of the load current I.sub.LOAD. As a result, the load current I.sub.LOAD may have the same shape as the gate control signal V.sub.GC as shown in FIG. **5**.

[0104] FIG. **6** shows examples of waveforms that illustrate an operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) non-conductive using a reverse phase-control dimming technique. A load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may use a phase control dimming technique to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **9**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half cycles may be the same as the operation of the first gate drive circuit in the positive half cycles as shown in FIG. **6**. Further, it should be appreciated that the slope of the dimmed-hot voltage V.sub.DHI when transitioning to zero as shown in FIG. **6** is exaggerated to illustrate the changes and correlations between the other signals shown in FIG. **6**, such as the drain voltage sense signal V.sub.DV, the turn-off Signal V.sub.T-OFF, the target signal V.sub.TRGT, the gate control signal V.sub.GC, and/or the load current I.sub.LOAD.

[0105] The gate drive circuit of the load control device may receive a drive signal V.sub.DR from a control circuit (e.g., the control circuit **120** and/or the control circuit **250**). The control circuit may drive the drive signal V.sub.DR high (e.g., towards the DC supply voltage V.sub.CC) at the beginning of a half-cycle of the AC mains line voltage. The gate drive circuit may generate a target signal V.sub.TRGT in response to the drive signal V.sub.DR from the control circuit. The control circuit may drive the drive signal V.sub.DR low at a control time t.sub.CNTL (e.g., a firing time). Prior to the control time t.sub.CNTL, the gate drive circuit may be configured to set the target signal V.sub.TRGT based on a turn-off signal V.sub.T-OFF (not shown) generated by a turn-off circuit (e.g., the turn-off circuit **228**) and a limit signal V.sub.LMT. The magnitude of the limit signal V.sub.LMT may range between a second magnitude (e.g., the supply voltage V.sub.CC) and a first magnitude (e.g., approximately circuit common). As noted herein, the control circuit may set the magnitude of the limit signal V.sub.LMT at the second magnitude or the first magnitude (e.g., or one or more values between the upper and first magnitudes) to control the magnitude of the maximum current limit I.sub.LMT of the load control device. In the waveforms of FIG. **6**, the magnitude limit signal V.sub.LMT is set to the second magnitude at the beginning of the half-cycle, and as such, a magnitude of the target signal V.sub.TRGT may be equal to the second limit-signal magnitude V.sub.MAG2. In response to the control circuit driving the magnitude of the drive signal V.sub.DR low at the control time t.sub.CNTL, the gate drive circuit may decrease the magnitude of the target signal V.sub.TRGT to a predetermined value, such as the first limit-signal magnitude V.sub.MAG1. The magnitude of the target signal V.sub.TRGT may be equal to the second limit-signal magnitude V.sub.MAG2 at the end of the turn-on time period T.sub.T-ON.

[0106] After decreasing the magnitude of the target voltage V.sub.TRGT from the second limit-signal magnitude V.sub.MAG2 to the first limit-signal magnitude V.sub.MAG1, the gate drive circuit may decrease (e.g., gradually decrease with respect to time) the magnitude of the target signal V.sub.TRGT while monitoring a characteristic of (e.g., a voltage developed across) the semiconductor switch. For example, the gate drive circuit may receive a drain voltage sense signal V.sub.DV from a drain voltage sense circuit (e.g., the drain voltage sense circuit **218**), where the drain voltage sense signal V.sub.DV indicates a magnitude of the voltage across the semiconductor switch. When semiconductor switch is conductive, the magnitude of the voltage across the semiconductor switch is substantially small (e.g., approximately zero volts). When the magnitude of the target voltage V.sub.TRGT decreases to a point that the semiconductor switch begins to become non-conductive, the magnitude of the voltage across the semiconductor switch may begin to increase causing the magnitude of the drain voltage sense signal V.sub.DV to increase as shown in FIG. **6**. The gate current sense circuit may determine when the magnitude of the drain voltage sense signal V.sub.DV exceeds a drain voltage threshold V.sub.TH (e.g., approximately 1V).

[0107] In some examples, the gate drive circuit may begin to shape the target signal V.sub.TRGT from a present value towards zero volts. For example, when the gate drive circuit detects that the voltage developed across the semiconductor switch has started to increase in magnitude such that the magnitude of the drain voltage sense signal V.sub.DV exceeds the drain voltage threshold V.sub.TH, the gate drive circuit may begin to shape the target signal V.sub.TRGT from a present value towards zero volts, for example, as shown in FIG. **6**. The gate drive circuit may shape the target signal V.sub.TRGT over a turn-off period T.sub.T-OFF (e.g., approximately 40 microseconds). In some examples, the transition period T.sub.TRAN may be equal to the turn-on time period T.sub.T-OFF, for example, when using reverse phase-control. The gate drive circuit may generate the target signal V.sub.TRGT using the turn off signal V.sub.T-OFF generated by the turn-off circuit. For example, the turn-off signal V.sub.T-OFF may have the same shape as the target voltage V.sub.TRGT during the turn-off period T.sub.T-OFF as shown in FIG. **6**. Accordingly, the gate drive circuit may use the drain voltage sense signal V.sub.DV as a trigger for the gate drive circuit to begin wave-shaping the turn-off signal V.sub.T-OFF (e.g., and also the target signal V.sub.TGRT). Alternatively, the gate drive circuit may wave-shape the target signal V.sub.TRGT irrespective of the magnitude of the drain voltage sense signal V.sub.DV. For example, the gate drive circuit may shape the target signal V.sub.TRGT after (e.g., immediately after) the magnitude of the target signal V.sub.TRGT is decreased from the second limit-signal magnitude V.sub.MAG2 to the first limit-signal magnitude V.sub.MAG1. The gate drive circuit may shape the turn-off signal V.sub.T-OFF (e.g., and also the target signal V.sub.TRGT) in a preconfigured shape, such as a S-shape, based on the magnitude of the drain voltage sense signal V.sub.DV. The gate drive circuit may shape the turn-off signal V.sub.T-OFF in a first shape during a first portion of the turn-off period T.sub.T-OFF, and shape the turn-off signal V.sub.T-OFF in a second shape during a second portion of the turn-off period T.sub.T-OFF (e.g., to create the S-shape).

[0108] The gate drive circuit may generate a gate control signal V.sub.GC based on the target signal V.sub.TRGT and a feedback signal V.sub.FB (not shown) that indicates a magnitude of a load current I.sub.LOAD conducted through a lighting load, as described herein. As described herein, the gate drive circuit may adjust a magnitude of the gate control signal V.sub.GC in response to a magnitude of the feedback signal V.sub.FB. For example, as shown in FIG. **6**, the gate drive circuit may adjust the magnitude of the gate control signal V.sub.GC in response to the magnitude of the feedback signal V.sub.FB, and thereby, adjust the magnitude of the load current I.sub.LOAD toward a target current. The gate drive circuit may render the semiconductor switch non-conductive by adjusting the magnitude of the gate control signal V.sub.GC to zero volts, and thereby, control the magnitude of the load current through the lighting load to zero amps. Since the gate drive circuit is using closed-loop control to adjust the magnitude of the gate control signal V.sub.GC based on the feedback signal V.sub.FB, the magnitude of the gate control signal V.sub.GC may indicate (e.g., be proportional to) the present magnitude of the load current I.sub.LOAD. As a result, the load current I.sub.LOAD may have the same shape as the gate control signal V.sub.GC as shown in FIG. **6**.

[0109] In some examples, the control circuit may control the overcurrent protection enable signal V.sub.OCP-EN to enable the overcurrent protection circuit at the zero-crossing of the AC mains line voltage at the beginning of the half-cycle. As noted herein, the overcurrent detection circuit may be configured to control the overcurrent detection signal V.sub.OCD to indicate the overcurrent condition after a trip time period T.sub.TRIP from when the overcurrent condition is first detected. The control circuit may be configured to enable the overcurrent protection circuit during the conduction period of each half-cycle. For example, when operating using the reverse phase-control technique, the control circuit may drive the magnitude of the OCP enable signal V.sub.OCP-EN high (e.g., toward the supply voltage V.sub.CC) at the zero-crossing at the beginning of the half-cycle of the AC mains line voltage. The control circuit may drive the magnitude of the OCP enable signal V.sub.OCP-EN low (e.g., toward circuit common) at the control time t.sub.CNTL. For example, the OCP enable signal V.sub.OCP-EN may be the same as to the limit signal V.sub.LMT as shown in FIG. **6**. As such, the control circuit may be configured to enable the overcurrent protection circuit during the conduction periods of each half-cycle.

[0110] FIG. **7** shows examples of waveforms that illustrate operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) conductive using a forward phase-control dimming technique. Itn the waveforms of FIG. **7**, the lighting load may be an inductive load. The load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may phase control the AC line voltage to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device for controlling a lighting load. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **7**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half-cycles may be the same as the operation of the first gate drive circuit in the positive half-cycles as shown in FIG. **7**.

[0111] During the operation of the gate drive circuit of FIG. **7**, the control circuit may determine that the lighting load is an inductive load. The control circuit may be configured to set a magnitude of a limit signal V.sub.LMT of the closed-loop gate drive circuit to the first magnitude (e.g., approximately circuit common) based on the lighting load being an inductive load. For example, the control circuit may set the magnitude of the limit signal V.sub.LMT to the first magnitude during the entire conduction period of a half-cycle of the AC mains line voltage. Since the magnitude of the limit signal V.sub.LMT is set to the first magnitude, the magnitude of the target signal V.sub.TRGT may be set to the first limit-signal magnitude V.sub.MAG1 at the end of the turn-on time period, which may set the magnitude of the maximum current limit I.sub.LMT to the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A). For instance, as described with respect to FIG. **5**, after the turn-on time period T.sub.T-ON, the gate drive circuit may be configured to set the magnitude of the target signal V.sub.TRGT based on the turn-on signal V.sub.T-ON and the limit signal V.sub.LMT.

[0112] Although the magnitude of the limit signal V.sub.LMT may range between the second magnitude and the first magnitude, the control circuit may set the magnitude of the limit signal V.sub.LMT to the first magnitude based on the lighting load being an inductive load. Since the lighting load is an inductive load, the lighting load may not need to conduct an inrush current as may be necessary by an incandescent or capacitive lighting load. Therefore, the control circuit may set the magnitude of the limit signal V.sub.LMT to the first magnitude during the entire conduction period of a half-cycle of the AC mains line voltage (e.g., without raising the limit signal V.sub.LMT to the second magnitude during the conductive time of the half-cycle). Further, maintaining the limit signal V.sub.LMT at the first magnitude during the half-cycle may ensure that the inductive load does not overheat and/or damage because the first magnitude will prevent a load current having a large magnitude from being conducted through the inductive load and/or may prevent an asymmetrical current having a large peak magnitude above a saturation threshold from being conducted by the load.

[0113] Accordingly, when using the forward phase-control dimming technique, the control circuit may set the magnitude of the limit signal V.sub.LMT to the first magnitude to adjust the magnitude of the target signal V.sub.TRGT to the first limit-signal magnitude V.sub.MAG1 to set the magnitude of the maximum current limit I.sub.LMT at a smaller magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A), to allow the overcurrent protection circuit to appropriately protect the inductive loads from large load currents.

[0114] FIG. **8** shows examples of waveforms that illustrate operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) conductive using a forward phase-control dimming technique. In the waveforms of FIG. **8**, the control circuit may be configured to adjust the limit signal V.sub.LMT during the conduction period of a half-cycle of the AC mains line voltage, for example, using a pulse. For example, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude (e.g., the supply voltage V.sub.CC) for a pulse period T.sub.PULSE that starts after the control time t.sub.CTNL1 (e.g., after the turn-on time T.sub.T-ON) and then adjust the limit signal V.sub.LMT to the first magnitude (e.g., approximately circuit common) for the rest of the conduction period of the half-cycle.

[0115] The load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may phase control the AC line voltage to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device for controlling a lighting load. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **8**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half-cycles may be the same as the operation of the first gate drive circuit in the positive half-cycles as shown in FIG. **8**.

[0116] During the operation of the gate drive circuit of FIG. **8**, the control circuit may set the magnitude of the limit signal V.sub.LMT of the closed-loop gate drive circuit to the second magnitude (e.g., the supply voltage V.sub.CC) for a pulse period T.sub.PULSE that starts after the control time t.sub.CTNL1 and then adjust the limit signal V.sub.LMT to the first magnitude (e.g., approximately circuit common) for the rest of the conduction period of the half-cycle. For example, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude (e.g., for the pulse period T.sub.PULSE after the turn-on time T.sub.T-ON and then adjust the limit signal V.sub.LMT to the first magnitude at the end of the turn-on time period T.sub.T-ON for the rest of the conduction period of the half-cycle. Accordingly, the target signal V.sub.TRGT may be set to the second limit-signal magnitude V.sub.MAG2 during the pulse period T.sub.PULSE and then set to the first limit-signal magnitude V.sub.MAG1 at the end of the pulse period T.sub.PULSE and for the remainder of the conduction period of the half-cycle of the AC mains line voltage. As such, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the second limit magnitude I.sub.MAG1 (e.g., approximately 20 A) during the pulse period

T.sub.PULSE, and then set the magnitude of the maximum current limit I.sub.LMT to the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A) at the end of the pulse period T.sub.PULSE and for the remainder of the conduction period of the half-cycle of the AC mains line voltage. For instance, as described with respect to FIG. **5**, after the turn-on time period T.sub.T-ON, the gate drive circuit may be configured to set the target signal V.sub.TRGT based on the turn-on signal V.sub.T-ON and the limit signal V.sub.LMT.

[0117] Therefore, by using the pulse period T.sub.PULSE (e.g., and setting the magnitude of the limit signal V.sub.LMT to the second magnitude) for a portion of the conduction period of the half-cycle of the AC mains line voltage, the control circuit is configured to allow the lighting load (e.g., should the lighting load be an incandescent load or a capacitive load) to conduct an inrush current during the pulse period T.sub.PULSE because the second magnitude will allow inrush current to be conducted through the lighting load. And, by setting the limit signal V.sub.LMT to the first magnitude after the pulse period T.sub.PULSE and for the rest of the conduction period of the half-cycle, the control circuit is configured to allow the lighting load (e.g., should the lighting load be an inductive load) to avoid overheating and/or damage due to saturation because the first magnitude will prevent a load current having a large magnitude from being conducted through the lighting load and/or may prevent an asymmetrical current having a large peak magnitude above a saturation threshold from being conducted by the load.

[0118] Accordingly, when using the forward phase-control dimming technique, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude during the pulse period T.sub.PULSE to adjust the magnitude of the target signal V.sub.TRGT to the second limit-signal magnitude V.sub.MAG2 to set the magnitude of the maximum current limit I.sub.LMT at a larger magnitude, such as the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A), during the pulse period T.sub.PULSE to allow for the lighting load to conduct a larger inrush current for a portion of the half-cycle, and set the magnitude of the limit signal V.sub.LMT to the first magnitude after the pulse period T.sub.PULSE and for the remainder of the half-cycle to adjust the magnitude of the target signal V.sub.TRGT to the first limit-signal magnitude V.sub.MAG1 to set the magnitude of the maximum current limit I.sub.LMT at a smaller magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A), to allow for the overcurrent protection circuit to appropriately protect the inductive loads from large load currents.

[0119] The length (e.g., duration) of the pulse period T.sub.PULSE may be adjustable (e.g., vary in duration). For example, the control circuit may use a pulse period T.sub.PULSE having a longer or shorter length during certain times and/or based on the load type. For example, in one example, the pulse period T.sub.PULSE may be as long as approximately 2,000 microseconds and as short as approximately 500 microseconds. Although other pulse periods T.sub.PULSE can be used. FIG. **9** shows examples of waveforms that illustrate operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) conductive using a forward phase-control dimming technique. In the waveforms of FIG. **9**, the control circuit may be configured to adjust the limit signal V.sub.LMT during the conduction period of a half-cycle of the AC mains line voltage, for example, by setting the magnitude of the limit signal V.sub.LMT to a different level for a period of time. For example, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude (e.g., the supply voltage V.sub.CC) for a pulse period T.sub.PULSE that starts after the control time t.sub.CTNL1 (e.g., after the turn-on time T.sub.T-ON) and then adjust the limit signal V.sub.LMT to the first magnitude (e.g., approximately circuit common) for the rest of the conduction period of the half-cycle.

[0120] In some examples, the control circuit may set a length of the pulse period T.sub.PULSE to a first time period T.sub.P1 (e.g., approximately 500 microseconds) to, for example, allow the capacitance of the capacitive load to appropriately charge each half-cycle or to a second time period T.sub.P2 (e.g., approximately 2000 microseconds) to, for example, allow an incandescent load to warm up. Further, in some examples, the control circuit may set the pulse period

T.sub.PULSE to the first time period T.sub.P1 or the second time period T.sub.P2 for a certain number of line cycles after powering on the lighting load (e.g., only a certain number of line cycles after powering on the lighting load). For instance, the control circuit may set the length of the pulse period T.sub.PULSE to the first time period T.sub.P1 or the second time period T.sub.P2 for a number of line cycles (e.g., approximately 120 half-cycles) after turning on the lighting load, and after the first number of line cycles, the control circuit may set the magnitude of the limit signal V.sub.LMT to the first magnitude for the entirety of the conduction periods of the half-cycles of future line cycles.

[0121] In some (e.g., alternative) examples, the control circuit may adjust the duration of the pulse period T.sub.PULSE during operation, such as when starting up or powering on the lighting load. For instance, the control circuit may set the pulse period T.sub.PULSE to the second time period T.sub.P2 for a first, initial number of line cycles of the AC mains line voltage (e.g., approximately 120 half-cycles), and then set the pulse period T.sub.PULSE to the first time period T.sub.P1 for a second, subsequent number of line cycles of the AC mains line voltage (e.g., forever), where the second time period T.sub.P2 may be greater than the first time period T.sub.P1 (e.g., approximately 2,000 microseconds and approximately 500 microseconds, respectively). Therefore, the control circuit may set the duration of the pulse period T.sub.PULSE to more than one value for a certain number of line cycles, for example, in response to powering on the lighting load (e.g., only a certain number of line cycles after powering on the lighting load).

[0122] The load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may phase control the AC line voltage to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device for controlling a lighting load. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **9**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half-cycles may be the same as the operation of the first gate drive circuit in the positive half-cycles as shown in FIG. **9**.

[0123] Accordingly, when using the forward phase-control dimming technique, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude during the pulse period T.sub.PULSE to adjust the magnitude of the target signal V.sub.TRGT to the second limit-signal magnitude V.sub.MAG2 to set the magnitude of the maximum current limit I.sub.LMT at a larger magnitude, such as the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A), during the pulse period T.sub.PULSE to allow for the lighting load to conduct a larger inrush current, and set the magnitude of the limit signal V.sub.LMT to the first magnitude after the pulse period T.sub.PULSE and for the remainder of the half-cycle to adjust the magnitude of the target signal V.sub.TRGT to the first limit-signal magnitude V.sub.MAG1 to set the magnitude of the maximum current limit I.sub.LMT at a smaller magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A), to allow for the overcurrent protection circuit to appropriately protect the inductive loads from large load currents. Further, the control circuit may adjust the duration or the time period T.sub.P of the pulse period T.sub.PULSE across multiple line cycles of the AC mains line voltage during operation, such as when starting up or powering on the lighting load, for example, by reducing the duration of the pulse period T.sub.PULSE from the second time period T.sub.P2 to the first time period T.sub.P1 (e.g., from approximately 2,000 μsec to approximately 500 μsec). Further, in some examples, after setting the pulse period T.sub.PULSE to the second time period T.sub.P2 for a number of line cycles, the control circuit may stop using the pulse period T.sub.PULSE altogether until the lighting load is turned off (e.g., the control circuit may reduce the limit signal V.sub.LMT to the lower value (e.g., approximately circuit common)). Thus, the control circuit may gradually transition the maximum current limit I.sub.LMT from a larger magnitude to a smaller magnitude (e.g., to accommodate various different types of lighting loads).

[0124] In some examples, the control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may include a digital-to-analog converter (DAC), and the control circuit of the control device may be configured to variably adjust (e.g., linearly adjust) the magnitude of the limit signal V.sub.LMT between the second magnitude (e.g., the supply voltage V.sub.CC) and the first magnitude (e.g., approximately circuit common) with respect to time during a conduction period of a half-cycle of the AC mains line voltage. For instance, the control circuit may set the magnitude of the limit signal V.sub.LMT using the digital-to-analog converter, and adjust the magnitude of the limit signal V.sub.LMT to adjust (e.g., linearly adjust) the magnitude of the target signal V.sub.TRGT between the second limit-signal magnitude V.sub.MAG2 and the first limit-signal magnitude V.sub.MAG1. As such, the control circuit may gradually transition the maximum current limit I.sub.LMT from a larger magnitude to a smaller magnitude (e.g., to accommodate various different types of lighting loads) during a half-cycles of the AC mains line voltage.

[0125] FIG. **10** illustrates examples of waveforms that illustrate the magnitude of the limit signal V.sub.LMT being adjusted between the second magnitude and the first magnitude with respect to time during a conduction period of a half-cycle of the AC mains line voltage. FIG. **10** shows examples of waveforms that illustrate operation of a gate drive circuit (e.g., the gate drive circuit **220**) to render a semiconductor switch (e.g., the FET Q**212**) conductive using a forward phase-control dimming technique. In the waveforms of FIG. **10**, the control circuit may be configured to adjust the limit signal V.sub.LMT during the conduction period of a half-cycle of the AC mains line voltage, for example, using a pulse. For example, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude (e.g., the supply voltage V.sub.CC) for a pulse period T.sub.PULSE that starts after the control time t.sub.CTNL1 (e.g., after the turn-on time T.sub.T-ON), adjust the magnitude of the limit signal V.sub.LMT from the second magnitude to the first magnitude (e.g., approximately circuit common or zero volts) across an adjustment period T.sub.ADJ, and then maintain the magnitude of the limit signal V.sub.LMT at the first magnitude for the rest of the conduction period of the half-cycle.

[0126] The load control device (e.g., the load control device **100** of FIG. **1** and/or the load control device **200** of FIG. **2**) may receive an AC mains line voltage from an AC power source and may phase control the AC line voltage to generate a dimmed-hot voltage V.sub.DH (e.g., a phase-control voltage) at a dimmed-hot terminal of the load control device for controlling a lighting load. The load control device may comprise two gate drive circuits with a first gate drive circuit (e.g., the first gate drive circuit **115** shown in FIG. **1**) controlling a first semiconductor switch (e.g., the first FET Q**111**) in the positive half-cycles (e.g., as shown in FIG. **10**) and a second gate drive circuit (e.g., the second gate drive circuit **116** shown in FIG. **1**) controlling a second semiconductor switch (e.g., the second FET Q**112**) in the negative half-cycles. The operation of the second gate drive circuit in the negative half-cycles may be the same as the operation of the first gate drive circuit in the positive half-cycles as shown in FIG. **10**.

[0127] As shown in FIG. **10**, the control device may linearly adjust the magnitude of the limit signal V.sub.LMT between the second magnitude and the first magnitude across the adjustment period T.sub.ADJ during a conduction period of a half-cycle of the AC mains line voltage. Although illustrated as a linear adjustment between the second magnitude and the first magnitude, the control circuit may variably adjust the limit signal V.sub.LMT using other shapes (e.g., multiple, discrete steps between the second magnitude and the first magnitude). Further, as illustrated in FIG. **10**, the control circuit may set the magnitude of the limit signal V.sub.LMT to the second magnitude for a pulse period T.sub.PULSE after the turn-on time period T.sub.T-ON. For example, the control circuit may set the length of the pulse period T.sub.PULSE to the first time period T.sub.P1 (e.g., approximately 500 microseconds). Then, after the pulse period T.sub.PULSE, the control circuit may linearly adjust the magnitude of the limit signal V.sub.LMT between the second magnitude and the first magnitude during the conduction period of the half-cycle of the AC mains line voltage. Then, the control circuit may keep the limit signal V.sub.LMT set to the first magnitude for the remainder of the duration of the conduction period of the half-cycle. Therefore, during the conduction period of a half-cycle of the AC mains line voltage, the control circuit may gradually transition the maximum current limit I.sub.LMT from a larger magnitude to a smaller magnitude across the adjustment period T.sub.ADJ (e.g., to accommodate various different types of lighting loads). The control circuit may perform the linear adjustment of the limit signal V.sub.LMT between the second magnitude and the first magnitude for a certain number of half-cycles of the AC mains line voltage (e.g., a certain number after turning on the lighting load), or the control circuit may perform the linear adjustment of the limit signal V.sub.LMT for all half-cycles of the AC mains line voltage.

[0128] Accordingly, when using the forward phase-control dimming technique, the control circuit may adjust the magnitude of the limit signal V.sub.LMT between the second magnitude and the first magnitude across the adjustment period T.sub.ADJ during the conduction period of a half-cycle of the AC mains line voltage to adjust (e.g., linearly adjust) the magnitude of the target signal V.sub.TRGT between the second limit-signal magnitude V.sub.MAG2 and the first limit-signal magnitude V.sub.MAG1, which in turn may adjust (e.g., linearly adjust) the maximum current limit I.sub.LMT between the larger magnitude, such as the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A), and the smaller magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A), across the adjustment period T.sub.ADJ during the conduction period of the half-cycle of the AC mains line voltage. Doing so might allow for the lighting load to conduct a larger inrush current during the pulse period T.sub.PULSE and during the adjustment period and also allow for the overcurrent protection circuit to appropriately protect the inductive loads from large load currents.

[0129] FIG. **11**A is a flowchart of an example procedure **1100** for setting the maximum current limit of a load control device based on the type of phase-control technique configured by the load control device. The procedure **1100** may be performed by a control circuit (e.g., the control circuit **120** and/or the control circuit **250**) of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) to control a drive circuit of the load control device (e.g., the gate drive circuit **115**, the gate drive circuit **116**, and/or the gate drive circuit **220**). The control circuit may execute the procedure **1100** at start-up, in response to a command to control the electrical load, and/or periodically. In some examples, the control circuit may execute the procedure **1100** in response to receiving a command to control power delivered to the lighting load. Alternatively or additionally, the control circuit may execute the procedure **1100** during every half-cycle of the AC power source.

[0130] The control circuit may execute the procedure **1100** to determine how to adjust the magnitude of the limit signal V.sub.LMT to control the maximum current limit I.sub.LMT. As described herein, the gate drive circuit may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT to control a semiconductor switch to control the power delivered to the lighting load. As also described herein, the gate drive circuit may limit the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch to the maximum current limit I.sub.LMT as indicated by the limit signal V.sub.LMT. As such, the magnitude of the limit signal V.sub.LMT may dictate the limit of the magnitude of the load current I.sub.LOAD (e.g., the maximum current level I.sub.LMT) conducted through the semiconductor switch. As described herein, the control circuit may generate the limit signal V.sub.LMT at a plurality of different magnitudes, such a first limit-signal magnitude V.sub.MAG1 (e.g., configured to adjust the maximum current limit I.sub.LMT to a first limit magnitude I.sub.MAG1) and a second limit-signal magnitude V.sub.MAG2 (e.g., configured to adjust the maximum current limit I.sub.LMT to a second limit magnitude I.sub.MAG2).

[0131] At **1102**, the control circuit may determine whether to control the lighting load using a forward phase-control technique. The control circuit may determine which technique to control the lighting load based on one or more factors. The control circuit may be configured (e.g., preconfigured) to use a specific phase-control technique. In some examples, the control circuit may determine the phase-control technique based on the type of lighting load that is coupled to the load control device. For example, the control circuit may determine that the lighting load is an inductive load when the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch (e.g., as indicated by the feedback voltage V.sub.FB) is out of phase with the AC mains line voltage V.sub.AC (e.g., if zero-crossings of the load current I.sub.LOAD conducted through the semiconductor switch do not line up with zero-crossings of the AC mains line voltage V.sub.AC as indicated by the zero-cross signal V.sub.ZC).

[0132] If the control circuit determines not to control the lighting load using a forward phase-control technique (e.g., determines to control the lighting load using a reverse phase-control technique) at **1102**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to a first level at **1104**. In some examples, the first level may be a second magnitude, such as the second limit magnitude I.sub.MAG2 (e.g., approximately 20 A). For instance, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the first level for the entire conduction period of each half-cycle where the load control device is controlling the power delivered to the electrical load. As described herein, the control circuit may set the magnitude of the limit signal V.sub.LMT to an upper value (e.g., to approximately the supply voltage V.sub.CC) to adjust the magnitude of the target signal V.sub.TRGT to the second limit-signal magnitude V.sub.MAG2, which in turn adjusts the maximum current limit I.sub.LMT to the first level. For example, as shown in the waveforms of FIG. **6**, the limit signal V.sub.LMT may set to the upper value at the end of the turn-on time period T.sub.T-ON, and as such, the target signal V.sub.TRGT may be equal to the second limit-signal magnitude V.sub.MAG2 at the end of the turn-on time period T.sub.T-ON. As such, if the control circuit is configured to use a phase-control technique other than forward phase-control (e.g., a reverse phase-control technique or a center phase-control technique), the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the first level.

[0133] If the control circuit determines to control the lighting load using a forward phase-control technique at **1102**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to a second level for a pulse period at **1106**. In some examples, the second level may be the second magnitude, such as the second limit magnitude I.sub.MAG2 (approximately 20 A). The control circuit may set the magnitude of the maximum current limit I.sub.LMT to the second level after the turn-on time T.sub.T-ON (e.g., which starts at the control time t.sub.CTNL1) and for the duration of the pulse period T.sub.PULSE. In some examples, the pulse period T.sub.PULSE may be approximately 500 μsec.

[0134] At **1108**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to a third level for the remaining of the conduction period of the half-cycle of the AC mains line voltage, before existing the procedure **1100**. In some examples, the third level may be equal to the first magnitude, such as the first limit magnitude I.sub.MAG1 (e.g., approximately 8 A). As such, the control circuit may adjust the maximum current limit I.sub.LMT from the first level to the second level at the end of the pulse period T.sub.PULSE, and maintain the maximum current limit I.sub.LMT at the second level for the remainder of the conduction period of the half-cycle of the AC mains line voltage. The control circuit may set the magnitude of the limit signal V.sub.LMT to the lower level (e.g., to approximately circuit common) to adjust the magnitude of the target signal V.sub.TRGT to a first limit-signal magnitude V.sub.MAG1, which in turn adjusts the maximum current limit I.sub.LMT to the third level. At the beginning of the next half-cycle of the AC mains line voltage, the control circuit may restart the procedure **1100** (e.g., or revert back to **1106**).

[0135] As such, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the third level at the end of the pulse period T.sub.PULSE and for the rest of the conduction period of the half-cycle of the AC mains line voltage. Therefore, using the procedure **1100**, when using a forward phase-control technique, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the second level for the pulse period T.sub.PULSE to allow the capacitance of the capacitive load to appropriately charge each half-cycle and/or an incandescent load to warm up, and set the magnitude of the maximum current limit I.sub.LMT to the third level at the end of the pulse period T.sub.PULSE and for the rest of the conduction period of the half-cycle to ensure that an inductive load does not conduct a load current having a peak magnitude above a saturation threshold and/or may prevent an asymmetrical current having a large peak magnitude above a saturation threshold from being conducted by the load.
[0136] FIG. **11**B is a flowchart of an example procedure **1120** for setting the maximum current limit of a load control device based on the type of phase-control technique configured by the load control device. The procedure **1120** may be performed by a control circuit (e.g., the control circuit **120** and/or the control circuit **250**) of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) to control a drive circuit of the load control device (e.g., the gate drive circuit **115**, the gate drive circuit **116**, and/or the gate drive circuit **220**). The control circuit may execute the procedure **1100** at start-up, in response to a command to control the electrical load, and/or periodically. In some examples, the control circuit may execute the procedure **1120** in response to receiving a command to control power delivered to the lighting load. Alternatively or additionally, the control circuit may execute the procedure **1120** during every half-cycle of the AC power source.
[0137] The control circuit may execute the procedure **1120** to determine how to control an over-current protection (OCP) circuit of the load control device (e.g., the first or second OCP circuits **117**, **118** of FIG. **1** and/or the OCP circuit **230** of FIG. **2**), how to control the drive signal V.sub.DR provided to a gate drive circuit of the load control device (e.g., the first or second gate drive circuit **115**, **116** of FIG. **1** and/or the gate drive circuit **220** of FIG. **2**), and how to control the magnitude of the limit signal V.sub.LMT to control the maximum current limit I.sub.LMT of the load control device.
[0138] At **1122**, the control circuit may detect a zero-crossing points of an AC mains line voltage V.sub.AC of a AC power source. For example, the load control device may comprise a zero-crossing detect circuit (e.g., the zero-cross detect circuit **130**) that may be coupled to the hot terminal H and the neutral terminal N of the load control device, and may generate a zero-cross signal V.sub.ZC that indicates the zero-crossing points of the AC mains line voltage V.sub.AC. As described herein, the control circuit may determine the time to render one or more semiconductor switches conductive and/or non-conductive based on the zero-crossing points of the AC mains line voltage.
[0139] At **1124**, the control circuit may determine whether to control the lighting load using a forward phase-control technique. The control circuit may determine which technique to control the lighting load based on one or more factors. The control circuit may be configured (e.g., preconfigured) to use a specific phase-control technique. In some examples, the control circuit may determine the phase-control technique based on the type of lighting load that is coupled to the load control device. For example, the control circuit may determine that the lighting load is an inductive load when the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch (e.g., as indicated by the feedback signal V.sub.FB) is out of phase with the AC mains line voltage V.sub.AC (e.g., if zero-crossings of the load current I.sub.LOAD as indicated by the feedback signal V.sub.FB do not line up with zero-crossings of the AC mains line voltage V.sub.AC as indicated by the zero-cross signal V.sub.ZC).
[0140] If the control circuit determines to control the lighting load using a forward phase-control technique at **1124**, the control circuit may wait for a control time t.sub.CNTL at **1126**. The control

circuit may be configured to adjust the control time (e.g., a firing time or a phase angle) each half-cycle of the dimmed-hot voltage V.sub.DH generated across the lighting load each half-cycle to control the amount of power delivered to the lighting load, and thus the present intensity level L.sub.PRES of the lighting load. For example, the control circuit may be configured to adjust the control time t.sub.CNTL each half-cycle to adjust the present intensity level L.sub.PRES of the lighting load towards a target intensity level L.sub.TRGT.

[0141] At **1128**, the control circuit may control a drive signal V.sub.DR that may be received by a gate drive circuit for controlling a semiconductor switch. The drive signal V.sub.DR may be provided to the gate drive circuit for adjusting a control time of the dimmed-hot voltage V.sub.DH generated across the lighting load and/or a magnitude of the load current I.sub.LOAD conducted through the lighting load, for example, to control the present intensity level L.sub.PRES of the lighting load. The control circuit may be configured to adjust a duty cycle (e.g., an on time) of the drive signal V.sub.DR to adjust the present intensity level L.sub.PRES of the lighting load towards a target intensity level L.sub.TRGT. When operating using a forward phase-control technique (e.g., as shown in FIGS. **5** and **7-10**), the control circuit may control the drive signal V.sub.DR to render the semiconductor switch non-conductive (e.g., control the magnitude of the drive signal V.sub.DR high towards the supply voltage V.sub.CC) at the beginning of the half-cycle, and then control the drive signal V.sub.DR to render the semiconductor switch conductive at the control time t.sub.CNTL.

[0142] At **1130**, the control circuit may wait a turn-on time period T.sub.T-ON (e.g., approximately 50 microseconds). In some examples, as described herein, the control circuit may shape the target signal V.sub.TRGT over the turn-on time period T.sub.T-ON to shape the target signal V.sub.TRGT. At **1132**, the control circuit may drive the OCP enable signal V.sub.OCP-EN high after the turn-on time period T.sub.T-ON to enable the OCP circuit of the load control device. The control circuit may drive the OCP enable signal V.sub.OCP-EN low (e.g., toward circuit common) at the next zero-crossing event. As such, the control circuit may be configured to enable the overcurrent protection circuit during the conduction periods of each half-cycle.

[0143] At **1134**, the control circuit may set the magnitude of the maximum current level I.sub.LMT to a first forward phase-control level. In some examples, the first forward phase-control level may be the second magnitude (e.g., approximately 20 A). As described herein, the gate drive circuit may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT to control a semiconductor switch to control the power delivered to the lighting load. As also described herein, the gate drive circuit may limit the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch to the maximum current limit I.sub.LMT as indicated by the limit signal V.sub.LMT. As such, the magnitude of the limit signal V.sub.LMT may dictate the maximum current level I.sub.LMT.

[0144] At **1136**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the forward phase-control level (e.g., the second magnitude) for the duration of the pulse period T.sub.PULSE. In some examples, the pulse period T.sub.PULSE may be approximately 500 μsec. Further, in some examples, the control circuit may adjust the duration of the pulse period T.sub.PULSE, for example, for a number of half-cycles of the AC mains line voltage after turning on the electrical load. For instance, the control circuit may be configured to set the pulse period T.sub.PULSE to a second time period T.sub.P2 (e.g., approximately 2,000 μsec) for a number of half-cycles (e.g., approximately 120 half-cycles) after turning on the electrical load, and after the number of half-cycles, change the pulse period T.sub.PULSE to a first time period T.sub.P1 (e.g., approximately 500 μsec) for subsequent half-cycles of the AC mains line voltage.

[0145] At **1138**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to a second forward phase-control level. In some examples, the second forward phase-control level may be the first magnitude (e.g., a first limit magnitude I.sub.MAG1, such as approximately 8 A). The control circuit may set the magnitude of the magnitude of the limit signal V.sub.LMT to the first magnitude (e.g., to approximately circuit common) to adjust the magnitude of the target signal V.sub.TRGT to a first limit-signal magnitude V.sub.MAG1, which in turn adjusts the maximum current limit I.sub.LMT to the first magnitude.

[0146] At **1140**, the control circuit may wait for the end of the half-cycle of the AC mains line voltage V.sub.AC. The control circuit may detect the end of the half-cycle of the AC mains line voltage V.sub.AC based on zero-crossing points of the AC mains line voltage V.sub.AC. At **1142**, the control circuit may disable the OCP circuit, for example, by driving the OCP enable signal V.sub.OCP-EN low at the zero-crossing event of the AC mains line voltage V.sub.AC. The control circuit may drive the OCP enable signal V.sub.OCP-EN low (e.g., toward circuit common) to disable the OCP circuit at the beginning of the next half-cycle of the AC mains line voltage V.sub.AC.

[0147] As such, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the second forward phase-control level at the end of the pulse period T.sub.PULSE and for the rest of the conduction period of the half-cycle of the AC mains line voltage. Therefore, using the procedure **1120**, when using a forward phase-control technique, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to the first forward phase-control level (e.g., second magnitude) for the pulse period T.sub.PULSE to allow the capacitance of the capacitive load to appropriately charge each half-cycle and/or an incandescent load to warm up, and set the magnitude of the maximum current limit I.sub.LMT to the second forward phase-control level (e.g., first magnitude) at the end of the pulse period T.sub.PULSE and for the rest of the conduction period of the half-cycle to ensure that an inductive load does not conduct a load current having a magnitude above a saturation threshold.

[0148] If the control circuit determines not to control the lighting load using a forward phase-control technique at **1124** (e.g., determine to control the lighting load using a reverse phase-control technique), the control circuit may enable the OCP circuit at **1144** (e.g., as described with respect to **1132**). At **1146**, the control circuit may set the magnitude of the maximum current level to the reverse phase-control level. In some examples, the reverse phase-control level may be the upper level (e.g., approximately 20 A). As described herein, the gate drive circuit may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT to control a semiconductor switch to control the power delivered to the lighting load. As also described herein, the gate drive circuit may limit the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch to the maximum current limit I.sub.LMT as indicated by the limit signal V.sub.LMT. As such, the magnitude of the limit signal V.sub.LMT may dictate the maximum current level I.sub.LMT.

[0149] At **1148**, the control circuit may wait for a control time t.sub.CNTL. The control circuit may be configured to adjust the control time t.sub.CNTL (e.g., a firing time or a phase angle) each half-cycle of the dimmed-hot voltage V.sub.DH generated across the lighting load each half-cycle to control the amount of power delivered to the lighting load, and thus the present intensity level L.sub.PRES of the lighting load. For example, the control circuit may be configured to adjust the control time t.sub.CNTL each half-cycle to adjust the present intensity level L.sub.PRES of the lighting load towards a target intensity level L.sub.TRGT.

[0150] At **1150**, the control circuit may control a drive signal V.sub.DR that may be received by a gate drive circuit for controlling a semiconductor switch. The drive signal V.sub.DR may be provided to the gate drive circuit for adjusting a control time of the dimmed-hot voltage V.sub.DH generated across the lighting load and/or a magnitude of the load current I.sub.LOAD conducted through the lighting load, for example, to control the present intensity level L.sub.PRES of the lighting load. The control circuit may be configured to adjust a duty cycle (e.g., an on time) of the drive signal V.sub.DR to adjust the present intensity level L.sub.PRES of the lighting load towards a target intensity level L.sub.TRGT. When operating using a reverse phase-control technique (e.g., as shown in FIG. **6**), the control circuit may control the drive signal V.sub.DR to render the semiconductor switch conductive (e.g., control the magnitude of the drive signal V.sub.DR high towards the supply voltage V.sub.CC) at the beginning of the half-cycle, and then control the drive signal V.sub.DR to render the semiconductor switch non-conductive at the control time t.sub.CNTL.

[0151] At **1152**, the control circuit may disable the OCP circuit, for example, by driving the OCP enable signal V.sub.OCP-EN low, and the procedure **1120** may exit. For example, the control circuit may drive the OCP enable signal V.sub.OCP-EN low at the beginning of a turn-off period T.sub.T-OFF. The control circuit may drive the OCP enable signal V.sub.OCP-EN high (e.g., toward circuit common) to enable the OCP circuit at the beginning of the next half-cycle of the AC mains line voltage V.sub.AC.

[0152] FIG. **11**C is a flowchart of an example procedure **1160** for adjusting a pulse period of a maximum current limit of a load control device in response to receiving a command to turn on the electrical load. The procedure **1160** may be performed by a control circuit (e.g., the control circuit **120** and/or the control circuit **250**) of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) to control a drive circuit of the load control device (e.g., the gate drive circuit **115**, the gate drive circuit **116**, and/or the gate drive circuit **220**). The control circuit may execute the procedure **1160** at start-up, in response to a command to control the electrical load, and/or periodically. For example, the control circuit may execute the procedure **1160** in response to a command to turn-on the electrical load. The control circuit may execute the procedure **1160** to adjust the pulse period of the maximum current limit at turn-on to accommodate different load types. The control device may perform the procedure **1160** when configured with a forward phase-control technique. The control circuit may perform the procedure **1160** in combination with other procedures described herein (e.g., the procedure **1100** and/or the procedure **1120**).

[0153] At **1162**, the control circuit may determine whether a turn-on command has been received. If a turn-on command has not been received, the control circuit may exit the procedure **1160**. If a turn-on command is received at **1162**, the control circuit may set the length (e.g., duration) of the pulse period T.sub.PULSE to a first value for a first period of time at **1164**. For example, as noted herein, the length (e.g., duration) of the pulse period T.sub.PULSE may be adjustable. For instance, the control circuit may set the pulse period T.sub.PULSE to a first time period T.sub.P1 for a first, initial number of line cycles of the AC mains line voltage (e.g., approximately 60 half-cycles). The first time period T.sub.P1 may be approximately 2,000 μsec. By setting the length of the pulse period T.sub.PULSE to a first value (e.g., approximately 2,000 μsec) for a period of time (e.g., approximately 120 half-cycles), the control circuit may allow the capacitance of the capacitive load to appropriately charge each half-cycle (e.g., allow for sufficient in-rush current) and/or an incandescent load to warm up over the first period of time.

[0154] At **1166**, the control circuit may set the length (e.g., duration) of the pulse period T.sub.PULSE to a second value (e.g., until the electrical load is turned off), and the procedure **1160** may exit. For example, the control circuit may set the pulse period T.sub.PULSE to a second time period T.sub.P2. The second time period T.sub.P2 may be approximately 500 μsec. By setting the length of the pulse period T.sub.PULSE to the second value (e.g., approximately 500 μsec) after the period of time, the control circuit may ensure that an inductive load does not conduct a load current having a magnitude above a saturation threshold.

[0155] In some examples, the control circuit may set the pulse period to the second value for a second, subsequent number of line cycles of the AC mains line voltage (e.g., approximately 60 half-cycles). In such examples, after the second, subsequent number of line cycles of the AC mains line voltage, the control circuit may be configured to reduce the limit signal V.sub.LMT to the lower value (e.g., approximately circuit common), which may adjust the magnitude of the target signal V.sub.TRGT to the first limit-signal magnitude V.sub.MAG1 for the line cycles of the AC mains line voltage after the use of the pulse period T.sub.PULSE. Therefore, the control circuit may set the length of the pulse period T.sub.PULSE to more than one value for a certain number of line cycles, for example, in response to powering on the lighting load (e.g., only a certain number of line cycles after powering on the lighting load).

[0156] FIG. **12** is a flowchart of an example procedure **1200** for controlling a maximum current limit of a load control device based on whether an electrical load coupled to the load control device is an inductive load. The procedure **1200** may be performed by a control circuit (e.g., the control circuit **120** and/or the control circuit **250**) of a load control device (e.g., the load control device **100** shown in FIG. **1** and/or the load control device **200** shown in FIG. **2**) to control a drive circuit of the load control device (e.g., the gate drive circuit **115**, the gate drive circuit **116**, and/or the gate drive circuit **220**). The control circuit may execute the procedure **1200** at start-up, in response to a command to control the electrical load, and/or periodically. The control circuit may execute the procedure **1200** to determine how to set the magnitude of the maximum current limit differently for inductive load types.

[0157] At **1202**, the control circuit may monitor one or more signals to determine if the electrical load is an inductive load. For example, the control circuit may be configured to determine that the load control device is connected to an inductive load if the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch (e.g., as indicated by the feedback voltage V.sub.FB) is out of phase with the AC mains line voltage V.sub.AC (e.g., if zero-crossings of the load current I.sub.LOAD as indicated by the feedback signal V.sub.FB do not line up with zero-crossings of the AC mains line voltage V.sub.AC as indicated by the zero-cross signal V.sub.ZC). If the control circuit determines that the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch is in of phase with the AC mains line voltage V.sub.AC, the control circuit may determine that it is not coupled to an inductive load.

[0158] If the control circuit determines that the load control device is connected to a load that is not an inductive load at **1204**, the control circuit may set the magnitude of the maximum current level I.sub.LMT to a non-inductive load level for the conduction period of the AC mains line voltage V.sub.AC at **1206**. In some examples, the non-inductive load level may be the second magnitude (e.g., approximately 20 A). As shown in FIG. **5**, the gate drive circuit may generate the target signal V.sub.TRGT based on the limit signal V.sub.LMT to control a semiconductor switch to control the power delivered to the lighting load. As also described herein, the gate drive circuit may limit the magnitude of the load current I.sub.LOAD conducted through the semiconductor switch to the maximum current limit I.sub.LMT as indicated by the limit signal V.sub.LMT. As such, the magnitude of the limit signal V.sub.LMT may dictate the maximum current level I.sub.LMT. Accordingly, when coupled to a load that is not an inductive load, such as an incandescent load or an capacitive load, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to a non-inductive load level to, for example, allow for the electrical load to conduct a larger inrush current for a portion of the half-cycle (e.g., in the case of a capacitive load) and/or allow the electrical load to warm up after turning on (e.g., in the case of an incandescent load).

[0159] If the control circuit determines that the load control device is connected to an inductive load at **1204**, the control circuit may set the magnitude of the maximum current limit I.sub.LMT to one or more inductive-load levels at **1208**. In some examples, the inductive-load levels may be a single level, such as the first magnitude (e.g., approximately 8 A), for example, as shown in FIG. **7**. In other examples, the inductive-load levels may be a combination of two different levels, such as the second magnitude and the first magnitude, for instance, as shown in FIGS. **8** and **9**. For instance, the inductive-load levels may include a first level that is used for a pulse period T.sub.PULSE, and a second level that is used for the remainder of the conduction period of the AC mains line voltage V.sub.AC. Further, in some examples, the inductive-load levels may include a plurality of different levels, such that the inductive-load levels change between a first level (e.g., the second magnitude) and a second level (e.g., the first magnitude) with respect to time during a conduction period of a half-cycle of the AC mains line voltage. In some instances, the inductive-load levels may change linearly between the first and second levels over an adjustment period, for

example, as shown in FIG. **10**. As noted herein, when connected to an inductive load, the control circuit may be configured with a forward phase-control technique.
[0160] Accordingly, when connected to an inductive load, the control circuit may be configured to set the magnitude of the maximum current limit I.sub.LMT at one or more inductive-load levels to allow the overcurrent protection circuit to appropriately protect the inductive loads from large load currents. For example, the one or more inductive-load levels may be set so as to prevent an inductive load from overheating or saturating when using a forward phase-control technique.

Claims

1. A load control device configured to control power delivered from an alternating-current (AC) power source to an electrical load, the load control device comprising: a controllably conductive device adapted to be coupled in series between the AC power source and the electrical load, wherein the AC power source is configured to generate an AC mains line voltage, and wherein the controllably conductive device comprises a semiconductor switch configured to conduct a load current through the electrical load; a closed-loop gate drive circuit coupled to the semiconductor switch and configured to render the semiconductor switch conductive or non-conductive at a control time during half-cycles of the AC mains line voltage, wherein the closed-loop gate drive circuit is configured to limit a magnitude of the load current conducted through the semiconductor switch to a maximum current limit; and a control circuit configured to: generate a drive signal to adjust the control time of the semiconductor switch during each half-cycle of the AC mains line voltage to control an amount of power delivered to the electrical load; control the drive signal to render the semiconductor switch conductive at a control time during a half-cycle of the AC mains line voltage to maintain the semiconductor switch conductive for a conduction period during the half-cycle of the AC mains line voltage; set the maximum current limit of the closed-loop gate drive circuit to a first limit for a period of time following the control time during the conduction period of the half-cycle of the AC mains line voltage; and set the maximum current limit of the closed-loop gate drive circuit to a second limit after an expiration of the period of time during the conduction period of the half-cycle of the AC mains line voltage, wherein the second limit is less than the first limit.
2. The load control device of claim 1, wherein the control circuit is configured to: set the maximum current limit to the first limit for a pulse period that starts at the control time during the half-cycle of the AC mains line voltage; and set the maximum current limit to the second limit at an end of the pulse period and during a remainer of the conduction period of the half-cycle of the AC mains line voltage.
3. The load control device of claim 2, wherein the pulse period is approximately 500 microseconds.
4. The load control device of claim 2, wherein the control circuit is configured to set the maximum current limit to the first limit for the pulse period to allow for a capacitance of a capacitive load to charge during the half-cycle of the AC mains line voltage.
5. The load control device of claim 2, wherein the control circuit is configured to: set the maximum current limit of the closed-loop gate drive circuit to the first limit for the pulse period following the control time of the semiconductor switch during an initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load; and after the initial plurality of half-cycles of the AC mains line voltage, set the maximum current limit of the closed-loop gate drive circuit to the second limit at the control time during subsequent half-cycles of the AC mains line voltage after the initial plurality of half-cycles and while controlling power to the electrical load.
6. The load control device of claim 5, wherein the control circuit is configured to shorten a duration of the pulse period over the initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load.
7. The load control device of claim 2, wherein the control circuit is configured to: set a length of

the pulse period to a second time period for a number of half-cycles after turning on the electrical load; and set the length of the pulse period to a first time period after the number of half-cycles and while controlling power to the electrical load.
8. The load control device of claim 7, wherein the first time period is approximately 500 microseconds, the second time period is approximately 2,000 microseconds, and the number of half-cycles is approximately 120 half-cycles.
9. The load control device of claim 2, wherein the control circuit is configured to: adjust the maximum current limit across a range of values from the first limit to the second limit over a plurality of half-cycles of the AC mains line voltage.
10. The load control device of claim 9, wherein the control circuit is configured to: linearly reduce the maximum current limit across the range of values from the first limit to the second limit over the plurality of half-cycles of the AC mains line voltage.
11. The load control device of claim 1, wherein, when operating in a forward phase-control technique, the control circuit is configured to: set the maximum current limit of the closed-loop gate drive circuit to the first limit for the period of time following the control time of the semiconductor switch during a conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load; and set the maximum current limit of the closed-loop gate drive circuit to the second limit after an expiration of the period of time during the conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load.
12. The load control device of claim 11, wherein, when operating in a reverse phase-control technique, the control circuit is configured to: set the maximum current limit of the closed-loop gate drive circuit to a third limit during the entire conduction period of each half-cycle of the AC mains line voltage while controlling power to the electrical load.
13. The load control device of claim 12, wherein the third limit is equal to the first limit.
14. The load control device of claim 1, wherein the closed-loop gate drive circuit is configured to: generate a target signal in response to the control circuit, receive a feedback signal indicative of a magnitude of the load current conducted through the semiconductor switch, and generate a gate control signal in response to the target signal and the feedback signal; wherein the semiconductor switch of the controllably conductive device is configured to be rendered conductive and non-conductive in response to the gate control signal.
15. The load control device of claim 1, wherein, when using a forward phase-control technique, the control circuit is configured to: control the drive signal to render the semiconductor switch non-conductive at a beginning of each half-cycle of the AC mains line voltage; and control the drive signal to render the semiconductor switch conductive at the control time during the each half-cycle of the AC mains line voltage and throughout a remainder of each half-cycle of the AC mains line voltage while controlling power to the electrical load.
16. The load control device of claim 15, wherein, when using a reverse phase-control technique, the control circuit is configured to: control the drive signal to render the semiconductor switches conductive at the beginning of each half-cycle of the AC mains line voltage; and control the drive signal to render the semiconductor switch non-conductive at the control time during each half-cycle of the AC mains line voltage.
17. The load control device of claim 1, wherein the control circuit is configured to control the semiconductor switch to be non-conductive for a non-conduction period and conductive for a conduction period during one or more half-cycles of the AC mains line voltage to control the amount of power delivered to the electrical load.
18.-**38**. (canceled)
39. A method performed by a load control device that is configured to control power delivered from an alternating-current (AC) power source to an electrical load, the method comprising: generating a drive signal to adjust a control time of a semiconductor switch of the load control device during

each half-cycle of a AC mains line voltage to control an amount of power delivered to the electrical load; controlling the drive signal to render the semiconductor switch conductive at a control time during a half-cycle of the AC mains line voltage to maintain the semiconductor switch conductive for a conduction period during the half-cycle of the AC mains line voltage; setting a maximum current limit of a closed-loop gate drive circuit to a first limit for a period of time following the control time during the conduction period of the half-cycle of the AC mains line voltage, wherein the closed-loop gate drive circuit is configured to limit a magnitude of the load current conducted through the semiconductor switch to the maximum current limit; and setting the maximum current limit of the closed-loop gate drive circuit to a second limit after an expiration of the period of time during the conduction period of the half-cycle of the AC mains line voltage, wherein the second limit is less than the first limit.

40. The method of claim 39, further comprising: setting the maximum current limit to the first limit for a pulse period that starts at the control time during the half-cycle of the AC mains line voltage; and setting the maximum current limit to the second limit at an end of the pulse period and during a remainer of the conduction period of the half-cycle of the AC mains line voltage.

41. (canceled)

42. (canceled)

43. The method of claim 40, further comprising: setting the maximum current limit of the closed-loop gate drive circuit to the first limit for the pulse period following the control time of the semiconductor switch during an initial plurality of half-cycles of the AC mains line voltage after turning on the electrical load; and after the initial plurality of half-cycles of the AC mains line voltage, setting the maximum current limit of the closed-loop gate drive circuit to the second limit at the control time during subsequent half-cycles of the AC mains line voltage after the initial plurality of half-cycles and while controlling power to the electrical load.

44.-**54**. (canceled)